

Acquiring Innovation under Information Frictions*

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Acquiring innovation through M&A is subject to information frictions, as assessing the value of innovative targets is a challenging task. We find an inverted U-shaped relation between firm innovation and takeover exposure; equity usage increases with target innovation; and the deal completion rate drops with innovation. We develop and estimate a model of acquiring innovation under information frictions, featuring endogenous merger, innovation, and offer composition decisions. Our estimates suggest that acquirers' due diligence reveals only 30% of private information possessed by targets. Eliminating information frictions increases capitalized merger gains by 59%, stimulates innovation, and boosts productivity, business dynamism, and social welfare. (*JEL* E20, G30, O40)

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Companies are buying innovation. As large companies need to be competitive and want to increase their footprints in a variety of different areas, one of the best ways to do that is through acquisition.

—Peter Levine and Andreessen Horowitz

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Innovation is the key driver of economic growth. Recent literature has documented that acquiring innovation and growth options is an important motive of corporate takeovers (e.g., Phillips and Zhdanov 2013; Bena and Li 2014; Levine 2017; Wang 2018). The existence of an active M&A market, in which innovation can be efficiently reallocated, incentivizes many firms to specialize in innovation with the anticipation of being acquired later on. These firms create significant value for their shareholders by exiting with high offer premiums. They also stimulate the growth of the whole economy by reallocating their innovation to more efficient users. Reallocating innovation through M&A, however, is subject to significant information frictions. Outsiders often find it challenging to assess the value and the impact of corporate innovation, and the adverse selection risk can arise as a major concern for acquirers who seek to purchase innovative targets. Despite the substantial yet largely separate literatures on information asymmetry, mergers and acquisitions, and corporate innovation, little has been done to understand the effect of information frictions on the reallocation of innovation through M&A. We aim to fill this gap in this paper.

We first document several novel findings regarding acquisitions of innovation. Using various measures that capture different aspects of corporate innovation, we document a robust inverted U-shaped relation between a firm's probability of being acquired and its stock of innovation. We also find that acquirers tend to use more equity in purchasing more innovative targets, and the probability of deal completion decreases with the target's innovativeness. These findings provide suggestive evidence for the role of information frictions in acquisitions of innovation. Specifically, as a firm becomes more innovative, its potential value as a target increases, which increases its exposure to takeovers. But meanwhile, a couple of counteracting forces are at play. First, a firm's stand-alone value is also increasing in its own innovation, which lifts the bar for the firm to accept an offer. Second, assessing the value of an innovative target becomes increasingly challenging when the target possesses more private information, and the adverse selection risk faced by potential acquirers makes them less willing to pay an attractive price, which hinders the chance of striking a successful deal. Taken together, these opposing effects contribute to an inverted U-shaped relation between a firm's exposure to takeovers and its innovativeness.

Using equity as payment can help mitigate the adverse selection risk, because bringing the target shareholders on board forces them to share the risk with acquirer shareholders (e.g., Hansen 1987; Fishman 1989; Eckbo, Giammarino, and Heinkel 1990). Acquirers, therefore, prefer using equity to purchase more innovative targets for hedging against the adverse selection risk. Equity usage, however, also can be costly to acquirers, because it is often accompanied by unfavorable market reactions to acquirer valuation due to the market's concern of the market timing hypothesis (e.g., Rhodes-Kropf, Robinson, and Viswanathan 2005; Rhodes-Kropf and Viswanathan 2004; Bhagat et al. 2005).

The trade-offs regarding the method of payment and acquirers' reluctance to pay an attractive price, both of which are driven by information frictions, reduce the deal completion rate when more innovative targets are involved.

Quantifying the effect of information frictions on the efficiency of innovation reallocation is challenging. We do not observe the acquirer's and the target's information sets, and, therefore, we cannot measure the adverse selection risk directly. More importantly, the M&A transactions and payment methods observed in the data are outcomes of an equilibrium in which acquirers and targets act strategically. To assess the effect of information frictions, we need to observe what would have happened in a parallel, counterfactual world in which information is perfect between acquirers and targets. Measuring this counterfactual is difficult because of the complexity in finding exogenous shocks that eliminate information frictions. Even if there were such a shock, it likely would be limited in scope, raising concerns about external validity. Overall, how to quantify the effect of information frictions without a model is unclear.

We overcome these challenges by estimating a general-equilibrium model of acquiring innovation under information frictions with endogenous productivity growth. In the model, acquirers can create value by purchasing innovative targets. However, they sometimes cannot perfectly observe the target firms' ongoing innovation. Instead, they form rational expectations over the true innovativeness based on the observables, and they make a take-it-or-leave-it offer to the targets. In the model, an acquirer has to decide on the amount of cash and equity to use in the offer, taking into account that the equity payment might be discounted. Upon receiving the offer, the target firm has to decide whether or not to accept it depending on its own information set and its continuation value. Firms in our model endogenously choose to become acquirers, targets, or remain stand-alone each period. In equilibrium, information frictions reduce acquirers' incentive to take risk and acquire targets with top-notch innovation, even though these targets can create the most value in M&A.

The model imposes no priors on whether information frictions are large, small, or even absent. We let the data tell us how important they are in transactions of innovation reallocation through M&A. We do so by estimating the model's parameters using the simulated method of moments (SMM). A few moments in the data are particularly important for our estimation. First, the inverted U-shaped relation between a firm's probability of being acquired and its stock of innovation, which we document in the data, reflects the two counteracting effects of innovation on firms' takeover exposure when information frictions are present. Second, acquirers endogenously choose the payment method by trading off the benefits of hedging against target value risk and the costs of equity payment being accepted at a discounted price. These patterns, therefore, help the model recover the extent of adverse selection risk faced by acquirers. The overall merger probability and the combined firm's

announcement returns are informative of merger gains, which in turn discipline our estimates of the merger technology.

Our estimates suggest that the information asymmetry between acquirers and targets is substantial. Specifically, we estimate that, on average, due diligence conducted by acquirers helps reveal only 30% of private information possessed by the targets, and thus, acquirers face severe adverse selection risk in purchasing innovative targets. Information frictions, therefore, create a substantial barrier to trades between acquirers and targets.

Using the estimated model as a laboratory, we perform several counterfactual analyses to gauge the effect of information frictions on firm innovation and asset reallocation. We find that eliminating asymmetric information between acquirers and targets has a strong, direct effect on value creation in the M&A market. As we decompose a firm's market value into its stand-alone value and the capitalized gains from M&A, we find that, with the estimated information frictions, the capitalized gains from the M&A market accounts for 6% of an average firm's market value, and eliminating information frictions in M&A boosts this ratio to 9.2%, representing an almost 60% increase. Part of the efficiency gains arise from speeding up the process of asset reallocation such that innovative firms are bought much earlier in their life cycle and valuable innovations are reallocated to better users at a much faster pace.

A more efficient M&A market also triggers long-term effects on firm innovation and overall productivity growth: as asset reallocation becomes more efficient, firms optimally choose to invest more on R&D and conduct more innovation, especially small firms who aim to be bought in the future. The largest firms, on the other hand, adopt a lower innovation intensity, as they have the option of improving their productivity by acquiring productive small firms rather than spending too much on in-house R&D. Overall, the average innovation investment and firm growth rate increase in the economy, reminiscent of the general equilibrium effects of reducing frictions in the secondary market for patented innovations found in Akcigit, Celik, and Greenwood (2016).

We also quantify the value of equity payment in helping mitigate the effect of information frictions. When we prevent firms from using equity as payment, we find that the capitalized gains from the M&A market decline from 6% of firm value to less than 4%, representing a 38% decrease. This finding suggests that, despite its cost, equity payment remains an important vehicle that acquirers use to mitigate the adverse selection risk arising from purchasing innovative targets.

Our paper contributes to three strands of literature. First, it adds to the growing literature that investigates the role of reallocating corporate innovation in mergers and acquisitions. For example, Sevilir and Tian (2011) and Bena and Li (2014) find that combining complementary innovation capabilities creates significant synergies in M&A; Higgins and Rodriguez (2006) and Phillips and Zhdanov (2013) show that large and mature firms often optimally outsource

their research and development by acquiring small and innovative firms; and Wang (2018) documents that entrepreneurs cater to potential acquirers' demand in innovation. The two papers most related to our work are Levine (2017) and David (2021). Levine (2017) builds a model in which acquirers pursue takeovers to obtain growth options from target firms. He shows that reallocation of growth options is another important motive of M&A beyond the traditional Q-theory of physical capital reallocation. David (2021) develops a search and matching model of M&A in general equilibrium, and uses it to evaluate the implications of merger activity for aggregate economic outcomes. The efficiency of the M&A market has been found to significantly contribute to the level of aggregate output and consumption. Our framework coincides with that of David (2021) in the synergistic merger technology and the use of a general equilibrium setting to assess the macroeconomic effects of the M&A market. The primary differences of our model are the endogenous productivity growth due to firm innovation, information frictions in the assessment of innovation that lead to adverse selection risk, and the use of equity versus cash in merger transactions to reduce this risk. Recent work by Cunningham, Ederer, and Ma (2021) documents that some firms acquire innovative targets to preempt future competition (i.e., killer acquisition), and their estimate suggests that about 6.4% of acquisitions in their sample are killer acquisitions. Our paper documents a few novel findings that help identify information asymmetry as an important friction in acquiring innovation, and we quantify the effects of such information frictions in the economy.

Our paper also pertains to the literature on information asymmetry and the method of payment in M&A. Theoretical work by Hansen (1987), Fishman (1989), and Eckbo, Giammarino, and Heinkel (1990) models the payment method as a vehicle to signal acquirers' valuation of targets and to hedge the adverse selection risk caused by target private information. More recently, a few studies show that overvalued acquirers are more likely to use equity as cheap currency to purchase real assets from targets, and this market timing hypothesis suggests that equity payment is often accepted at a discounted price.¹ Vladimirov (2015) further argues that how cash bids are financed is equally important as the method of payment, because when acquirers lack access to competitive financing, they tend to finance their cash bids with equity and that often leads to underbidding and lower takeover premiums. Our model incorporates the method of payment as acquirers' endogenous choice, and our estimates show that equity usage helps mitigate information frictions in acquiring innovation, but meanwhile it cannot fully insure acquirers from the adverse selection risk due to its cost to acquirers.

Methodologically, our paper belongs to the growing literature that employs structural model calibration or estimation to answer standard corporate

¹ See, for example, Shleifer and Vishny (2003), Rhodes-Kropf and Viswanathan (2004), Rhodes-Kropf, Robinson, and Viswanathan (2005), Bhagat et al. (2005), and Li, Taylor, and Wang (2018).

finance questions in capital investment, leverage choice, CEO turnover, market competition and valuation, and particularly M&As, as summarized in Strebulaev and Whited (2012). In the field of M&As, Warusawitharana (2008) and Yang (2008) construct and estimate models that link asset purchases and sales to firms' productivity, and their findings are consistent with the neoclassical Q-theory of M&As. Gorbenko and Malenko (2014) estimate valuations of strategic and financial bidders using a hand-collected data set of preannouncement bids and conclude that different targets appeal to different types of bidders. Albuquerque and Schroth (2010, 2015) estimate structural models to quantify the private benefits of control in block trades. Eisfeldt and Rampini (2008) develop a dynamic model to study the effect of agency problem on capital reallocation over the business cycle. Dimopoulos and Sacchetto (2017) develop and calibrate a dynamic industry-equilibrium model to quantify the impact of merger activity on productive efficiency. Our paper contributes to this literature by estimating the effect of information frictions in acquisitions of innovation.

1. Suggestive Evidence

To motivate the pivotal role of information frictions in our model and estimation, in this section, we present a few empirical findings regarding the relation between mergers and acquisitions and firm innovation. They provide suggestive evidence on how information frictions affect firms' involvement in the M&A market and their M&A outcomes.

We first examine the relation between a firm's probability of being acquired and its innovation stock. We gauge a firm's innovation using several measures that capture different aspects of innovation, including the total number of patents, patent citations, patent originality, and breakthrough innovation (i.e., tail innovations). Table 1 defines the variables.

We start with the univariate analysis in which we simply plot a firm's probability of being acquired against its innovation without any controls. Figure 1 shows the results for different innovation measures. There exists a robust hump-shaped relation between a firm's probability of being acquired and its innovation across all measures.

We further confirm our findings using multivariate regressions in which we control for a wide range of variables that are documented in previous studies to affect a firm's takeover exposure. Specifically, we perform the following regression analysis:

$$I_{i,t+1}^{acquired} = A_1 + B_1 \cdot Innov_{i,t} + C_1 \cdot Innov_{i,t}^2 + Q_1 \cdot X_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $I_{i,t+1}^{acquired}$ is a dummy that equals one if a firm i is acquired in year $t+1$, $Innov_{i,t}$ is one of the innovation measures for firm i in year t , $X_{i,t}$ is a vector of control variables including firm size, leverage, market-to-book ratio, return on assets, cash holding, R&D expenditure, asset tangibility, dividend payment

Table 1
Variable definitions

Notation	Variable	Definition
<i>Patent count</i>	Innovation quantity (patent count)	$\ln(1 + \text{patcount})$
<i>Citations</i>	Innovation quality (patent citations)	$\ln(1 + \text{citation})$
<i>Originality</i>	Innovation originality	$\ln(1 + \text{originality})$
<i>Breakthrough</i>	Breakthrough innovation	$\ln(1 + \text{tail innov})$
<i>Size</i>	Logarithm of market equity	$\ln(ME)$
<i>Leverage</i>	Market leverage	$\frac{dlrt+dlc}{dlrt+dlc+ME+PSLV-txditc}$
<i>MB</i>	Market-to-book equity ratio	$\frac{ME}{BE}$
<i>ROA</i>	Return on assets	$\frac{ni}{at}$
<i>Cash</i>	Cash holdings	$\frac{che}{at}$
<i>R&D</i>	Research and development	$\frac{xrd}{at}$
<i>Tangibility</i>	Asset tangibility	$\frac{ppent}{at}$
<i>Div</i>	Dividend dummy	$I_{div>0}$
<i>Age</i>	Firm age	age
<i>RRV</i>	Misvaluation measure	As in Rhodes-Kropf, Robinson, and Viswanathan (2005)
<i>Diversification</i>	Diversifying dummy	$I_{sicacq \neq sic_{tar}}$
<i>RelSize_{tar}</i>	Target size relative to the acquirer	$\frac{ME_{tar}}{ME_{acq}}$
<i>PrOfStk</i>	Fraction of equity in bids	$\frac{Equity\ payment}{Total\ payment}$
<i>ln(Assets)</i>	Logarithm of assets	$\ln(at)$
<i>CombRet</i>	Combined firm abnormal return	Combined firm 3-day CAR plus 30-day run-up

This table defines the variables used in the paper. The sample consists of all U.S.-listed firms, covering the period of 1980–2006. All innovation variables are constructed as the stock of past innovations following the perpetual inventory method with a depreciation rate of 6%. Patent count is the number of patents applied for by a firm in a given year. Patent citations are total citations received by patents applied for by a firm in a given year, which are corrected for truncation and technology class biases following Hall, Jaffe, and Trajtenberg (2001). Originality is the dispersion of technology classes cited by the firm's patents as described in Hall, Jaffe, and Trajtenberg (2001). Breakthrough innovations are defined as the number of patents that are in the top 10% of all patents according to the number of citations received among all patents applied for in that year as in Acemoglu, Kacigit, and Celik (Forthcoming).

dummy, firm age, the square of firm age, year fixed effects, and industry fixed effects. We include both $Innov_{i,t}$ and its quadratic term $Innov_{i,t}^2$ to capture the inverted U-shaped relation between a firm's probability of being acquired and its innovation. Table 2 presents the results. We find that even after controlling for a wide range of firm characteristics, the inverted U-shaped relation is still robust in the sense that for all innovation measures we use, we find a positive coefficient for the innovation measure and a negative coefficient for its quadratic term, both of which are statistically significant at the 1% level in most regression specifications. The turning point, calculated as $innov^* = -\frac{B_1}{2C_1}$, also lies around the 50th to 75th percentiles of each innovation measure, indicating that the inverted U-shaped relation is empirically relevant.² Taken together, we find that

² We also perform a more rigorous hypothesis test for the existence of an inverted U shape, following Lind and Mehlum (2010). Table E4 in the Internet Appendix displays the results, which confirm the presence of an inverted-U relation between takeover exposure and our measures of innovation. See more details in Section D of the Internet Appendix.

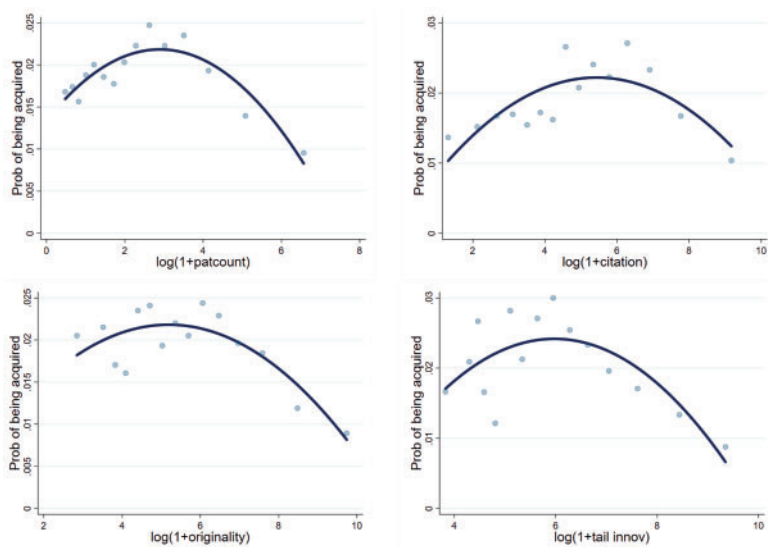


Figure 1
Firm innovation and the probability of being acquired

This figure illustrates the relation between a firm’s past innovation stock and its probability of being acquired next year without any controls. The measures of firm innovation are patent count, patent citations, originality, and tail innovations (top 10%). Variables are defined in Table 1. Dots represent the average probability of being acquired for firms in intervals partitioned by their innovation, and the curves represent the fitted value generated by quadratic regressions.

firms with an intermediate level of innovation are the most likely to be acquired, while firms with very weak or very strong innovation tend to stand alone. The inverted U-shaped relation is consistent with two counteracting forces brought about by firm innovation. On the one hand, innovative firms are more attractive targets in the M&A market, because they have the potential to create more value; on the other hand, if acquirers have imperfect information regarding target firm innovation, the adverse selection risk induced by information frictions makes it quite challenging to acquire highly innovative firms.

Next, we investigate the payment method in acquisitions of innovation. Previous studies have shown that the method of payment also can be influenced by information asymmetry (see, e.g., Hansen 1987; Fishman 1989; Eckbo, Giammarino, and Heinkel 1990). Specifically, if an acquirer is more concerned with information asymmetry about target quality, it may prefer using equity to reduce the adverse selection risk. This is because using equity brings the target shareholders on board, and, therefore, any losses arising from adverse selection are shared between the acquirer shareholders and target shareholders. Instead, cash bids allow the target shareholders to cash out and the acquirer shareholders have to shoulder all adverse selection costs. We perform the following regression analysis regarding the payment method:

$$PrcOfStk_{i,t+1} = A_2 + B_2 \cdot Innov_{i,t} + q_2 \cdot X_{i,t} + \varepsilon_{i,t}, \tag{2}$$

Table 2
Firm innovation and the probability of being acquired

Dependent var:		Probability of being acquired (annual)			
Innovation var:	Patent count	Citations	Originality	Breakthrough	
<i>Innov</i>	0.0036** (2.39)	0.0060*** (3.57)	0.0061*** (2.91)	0.0179*** (4.50)	
<i>Innov</i> ²	−0.0009*** (−4.46)	−0.0007*** (−4.12)	−0.0007*** (−4.18)	−0.0016*** (−5.03)	
<i>ln(Assets)</i>	0.0025*** (4.03)	0.0020*** (4.09)	0.0026*** (3.79)	0.0018*** (2.69)	
<i>Leverage</i>	−0.0065*** (−3.01)	−0.0060*** (−2.80)	−0.0054* (−1.93)	−0.005 (−1.46)	
<i>MB</i>	−0.0001 (−0.51)	−0.0001 (−0.50)	−0.0001 (−0.86)	−0.0001 (−0.86)	
<i>ROA</i>	0.0015** (2.47)	0.0018** (2.50)	0.0019*** (3.49)	0.0033*** (2.89)	
<i>Cash</i>	−0.0001 (−0.02)	0.0001 (0.01)	−0.0003 (−0.05)	0.0043 (0.54)	
<i>R&D</i>	0.0182*** (5.23)	0.0174*** (4.28)	0.0181*** (5.32)	0.0183** (2.37)	
<i>Tangibility</i>	−0.006 (−1.56)	−0.0053 (−1.34)	−0.0072 (−1.68)	−0.0061 (−1.07)	
<i>Div</i>	−0.0080*** (−3.73)	−0.0078*** (−3.68)	−0.0085*** (−3.42)	−0.0085*** (−3.04)	
<i>Age</i>	0.0010*** (3.92)	0.0010*** (3.93)	0.0010*** (3.98)	0.0012*** (3.79)	
<i>Age</i> ²	−0.0000*** (−3.35)	−0.0000*** (−3.47)	−0.0000*** (−3.29)	−0.0000*** (−3.18)	
<i>Const</i>	0.0085** (2.51)	0.0012 (0.20)	−0.0001 (−0.01)	−0.0419*** (−3.23)	
Industry FE	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	
N	59,822	57,885	49,348	35,433	
Adjusted R ²	.006	.005	.006	.007	

This table reports the results from Equation (1). *Innov* is the firm’s innovation, measured by its patent count (column 1), patent citations (column 2), originality (column 3), and breakthrough innovations (column 4). All innovation variables are constructed as the stock of past innovations following the perpetual inventory method with a depreciation rate of 6%. Patent count is the number of patents applied for by a firm in a given year. Patent citations are total citations received by patents applied for by a firm in a given year, which are corrected for truncation and technology class biases following Hall, Jaffe, and Trajtenberg (2001). Originality is the dispersion of technology classes cited by the firm’s patents as described in Hall, Jaffe, and Trajtenberg (2001). Breakthrough innovations are defined as the number of patents that are in the top 10% of all patents according to the number of citations received among all patents applied for in that year as in Acemoglu, Akcigit, and Celik (Forthcoming). See Table 1 for other variable definitions. Standard errors are clustered by the firm’s industry (Fama-French 48-industry classification).

where $PrcOfStk_{i,t+1}$ is the percentage of equity usage in a bid, $Innov_{i,t}$ is the target’s innovation, and $X_{i,t}$ is a vector of control variables, including acquirer innovation, acquirer and target misvaluation measures created by Rhodes-Kropf, Robinson, and Viswanathan (2005), acquirer and target size, market-to-book ratio, return on assets, leverage, dummy for diversification merger, year fixed effects, and industry fixed effects. Table 3 reports the results.

We find that equity usage indeed increases with the target’s innovation in general, consistent with the argument that acquirers prefer using equity as payment to mitigate the adverse selection risk caused by information asymmetry regarding the target’s innovation quality. We also find that equity usage increases with the acquirer’s overvaluation, consistent with the market

Table 3
Firm innovation and the method of payment

Dependent var:	Equity share in acquisition offer			
Innovation var:	Patent count	Citations	Originality	Breakthrough
<i>Innov_{tar}</i>	0.0237 (1.64)	0.0243** (2.18)	0.0265* (1.80)	0.0425* (1.94)
<i>RRV_{acq}</i>	0.1060*** (3.36)	0.1070*** (3.65)	0.0916*** (3.02)	0.1138*** (3.88)
<i>Innov_{acq}</i>	0.0021 (0.22)	0.0019 (0.30)	-0.0077 (-1.16)	0.0074 (0.83)
<i>Size_{acq}</i>	-0.0646*** (-4.74)	-0.0684*** (-5.34)	-0.0489*** (-4.89)	-0.0776*** (-3.36)
<i>MB_{acq}</i>	0.0182 (1.62)	0.0212* (1.75)	0.0183 (1.53)	0.0002 (0.01)
<i>Leverage_{acq}</i>	-0.0082 (-0.08)	0.0025 (0.02)	-0.0468 (-0.42)	-0.1368 (-1.22)
<i>ROA_{acq}</i>	-0.3549*** (-2.84)	-0.3668*** (-2.91)	-0.3428*** (-3.06)	-0.3272 (-1.45)
<i>RRV_{tar}</i>	0.0923*** (3.87)	0.0961*** (4.05)	0.0898*** (4.13)	0.1197*** (3.96)
<i>RelSize_{tar}</i>	0.0031 (0.27)	0.0029 (0.26)	0.0094 (1.23)	-0.0003 (-0.01)
<i>MB_{tar}</i>	0.0079 (0.71)	0.0028 (0.22)	0.0083 (0.82)	-0.0026 (-0.20)
<i>Leverage_{tar}</i>	0.1192 (1.30)	0.1169 (1.24)	0.1027 (1.04)	0.0710 (0.77)
<i>ROA_{tar}</i>	0.0375 (0.44)	0.0302 (0.39)	0.0047 (0.06)	0.0901 (1.16)
<i>Diversification</i>	-0.0255 (-0.51)	-0.0190 (-0.38)	0.0158 (0.26)	-0.0226 (-0.40)
<i>Const</i>	1.4836*** (5.30)	1.4976*** (5.66)	1.2204*** (5.53)	1.6832*** (3.33)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	605	590	530	410
Adjusted R^2	.253	.255	.245	.279

This table reports the results from Equation (2). *Innov* is the firm's innovation, measured by its patent count (column 1), patent citations (column 2), originality (column 3), and breakthrough innovations (column 4). All innovation variables are constructed as the stock of past innovations following the perpetual inventory method with a depreciation rate of 6%. Patent count is the number of patents applied for by a firm in a given year. Patent citations are total citations received by patents applied for by a firm in a given year, which are corrected for truncation and technology class biases following Hall, Jaffe, and Trajtenberg (2001). Originality is the dispersion of technology classes cited by the firm's patents as described in Hall, Jaffe, and Trajtenberg (2001). Breakthrough innovations are defined as the number of patents that are in the top 10% of all patents according to the number of citations received among all patents applied for in that year as in Acemoglu, Akcigit, and Celik (Forthcoming). Table 1 defines all other variables. Standard errors are clustered by the acquirer's industry (Fama-French 48-industry classification).

timing hypothesis. This analysis, therefore, highlights the potential costs and benefits of equity usage. On the one hand, when acquirers are uncertain about the targets' innovation quality, they prefer using equity as payment to hedge the adverse selection risk. On the other hand, equity usage may trigger investors' concern of market timing, in turn leading to unfavorable market reaction to acquirers' valuation. These two counteracting forces determine the optimal choice of offer composition in merger transactions.

Last, we look at how the deal completion rate varies with the target's innovativeness. Since equity usage is costly and cannot eliminate the adverse

Table 4
Firm innovation and the probability of deal completion

Dependent var:		Offer acceptance indicator			
Innovation var:	Patent count	Citations	Originality	Breakthrough	
<i>Innov_{tar}</i>	−0.0362*** (−3.66)	−0.0295*** (−3.64)	−0.0287*** (−3.04)	−0.0441*** (−3.32)	
<i>RRV_{acq}</i>	−0.0203 (−0.84)	−0.0190 (−0.78)	−0.0064 (−0.25)	−0.0305 (−0.98)	
<i>Innov_{acq}</i>	0.0030 (0.36)	0.0029 (0.49)	0.0041 (0.71)	−0.0054 (−0.83)	
<i>Size_{acq}</i>	0.0294** (2.30)	0.0304** (2.52)	0.0293** (2.42)	0.0332** (2.16)	
<i>MB_{acq}</i>	0.0035 (0.38)	0.0042 (0.45)	0.0089 (0.91)	−0.0001 (−0.00)	
<i>Leverage_{acq}</i>	−0.0999 (−1.31)	−0.0992 (−1.28)	−0.0737 (−0.92)	−0.1720 (−1.60)	
<i>ROA_{acq}</i>	−0.1498 (−1.35)	−0.1499 (−1.35)	−0.0975 (−0.86)	−0.1238 (−0.78)	
<i>RRV_{tar}</i>	0.0398** (2.07)	0.0412** (2.10)	0.0249 (1.21)	0.0719*** (3.15)	
<i>RelSize_{tar}</i>	−0.0559*** (−4.11)	−0.0570*** (−4.19)	−0.0456*** (−3.09)	−0.0789** (−2.44)	
<i>MB_{tar}</i>	−0.0190** (−2.23)	−0.0199** (−2.26)	−0.0134 (−1.43)	−0.0234** (−2.28)	
<i>Leverage_{tar}</i>	−0.0097 (−0.16)	−0.0161 (−0.26)	−0.0525 (−0.81)	−0.0134 (−0.18)	
<i>ROA_{tar}</i>	−0.1073 (−1.46)	−0.1235* (−1.66)	−0.1583* (−1.95)	−0.0727 (−0.83)	
<i>Diversification</i>	−0.0223 (−0.75)	−0.0251 (−0.83)	−0.0246 (−0.77)	0.0036 (0.10)	
<i>Const</i>	0.5974* (1.75)	0.6178* (1.86)	0.5081 (1.24)	0.7062* (1.86)	
Industry FE	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	
N	678	663	577	457	
Adjusted R ²	.136	.134	.135	.146	

This table reports the results from Equation (3). *Innov* is the firm's innovation, measured by its patent count (column 1), patent citations (column 2), originality (column 3), and breakthrough innovations (column 4). All innovation variables are constructed as the stock of past innovations following the perpetual inventory method with a depreciation rate of 6%. Patent count is the number of patents applied for by a firm in a given year. Patent citations are total citations received by patents applied for by a firm in a given year, which are corrected for truncation and technology class biases following Hall, Jaffe, and Trajtenberg (2001). Originality is the dispersion of technology classes cited by the firm's patents as described in Hall, Jaffe, and Trajtenberg (2001). Breakthrough innovations are defined as the number of patents in the top 10% of all patents according to the number of citations received among all patents applied for in that year as in Acemoglu, Akcigit, and Celik (Forthcoming). Table 1 defines all other variables. Standard errors are clustered by the acquirer's industry (Fama-French 48-industry classification).

selection risk, deal completion rate is expected to drop as information frictions intensify. Specifically, we perform the following regression analysis:

$$I_{i,t+1}^{complt} = A_3 + B_3 \cdot Innov_{i,t} + \varrho_3 \cdot X_{i,t} + \varepsilon_{i,t}, \quad (3)$$

where $I_{i,t+1}^{complt}$ is the dummy that the deal completes, conditioning on firm i receiving a bid. The independent variables are defined in Equation (2). In Table 4, we find that the probability of deal completion significantly decreases with the target's innovation across all four measures.

2. Model Setup

Motivated by the evidence presented above, we now construct a model of acquiring innovation with information frictions. The model endogenizes firms' innovation and M&A decisions, and the method of payment used in these transactions.

In the model, time is discrete and denoted by $t \in \{0, 1, 2, \dots\}$. There is a continuum of firms, indexed by $i \in [0, 1]$, which are heterogeneous in productivity $z_{it} > 0$. Firms use capital to produce a homogeneous final consumption good. We specify the production function in Section 2.2 below. As we will show later, decreasing returns to scale implies that a firm's flow profit is increasing in its productivity z_{it} . Firms invest in costly R&D to maintain and improve their productivity z_{it} over time.

Firms can also participate in a merger market where they act as an acquirer or a target. The M&A market features random matching: each period, a firm meets with a potential merging partner, and they decide whether to merge together. We assume that firms' contemporaneous productivity is privately observed, but conditional on a matching, with some probability, the acquirer firm can observe the contemporaneous productivity of the target through due diligence. This probability captures the information friction in our model. If the contemporaneous productivity is not revealed, the acquirer forms a rational expectation based on the target's productivity in previous periods.

The acquirer then makes a take-it-or-leave-it offer to the target firm and specifies the method of payment (cash, equity, or a mixed offer). After receiving the offer, the target firm decides whether or not to accept it. This decision hinges on the future opportunities available to the target as well as the expected offer value perceived by the target. If the merger completes, two firms merge into one combined firm, and the target firm ceases to exist. We also assume an exogenous rate of exit of all firms in the model. All exiting firms (due to endogenous acquisition or exogenous exit) are replaced by new firms such that the total measure of firms remains constant.

2.1 Household

We assume a representative household with time-separable preferences exhibiting constant relative-risk aversion. The time discount rate is $\beta \in (0, 1)$. The solution to the standard household optimization problem delivers the Euler equation $r = \beta^{-1} - 1$ in a stationary equilibrium, which governs the risk-free discount rate, r , in our model.

2.2 Firm production and innovation

Each period, firms produce a homogeneous final consumption good in our model with the Cobb-Douglas production function:

$$y_{it} = z_{it}^{\zeta} k_{it}^{\kappa}, \quad (4)$$

where z_{it} is the firm-specific productivity, and k_{it} is the capital input. We impose $\zeta \in (0, 1)$, $\kappa \in (0, 1)$, and $\zeta + \kappa = 1$. Since $\kappa < 1$, the production function features decreasing returns to scale.

We assume that there is no adjustment cost of capital, and the capital depreciation rate is $\delta \in (0, 1)$. A firm chooses optimal capital, within each period, to maximize its period profits:

$$\Pi_t(z_{it}) = \max_{k_{it} \geq 0} \left\{ z_{it}^\zeta k_{it}^\kappa - (r + \delta)k_{it} \right\}, \quad (5)$$

Solving the profit maximization problem above yields the firm's flow profit as follows:

$$\Pi_t(z_{it}) = \pi z_{it}, \quad (6)$$

where $\pi \equiv \zeta \left(\frac{\kappa}{r + \delta} \right)^{\frac{\kappa}{1 - \kappa}}$ is a time-invariant constant in a stationary equilibrium. The firm's period profits are therefore linear in its productivity.

Firm productivity z_{it} fluctuates over time, and firms innovate to maintain and improve their productivity. Specifically, we model firm productivity as an AR(1) process:

$$\log z_{it+1} = \mu_{it} + \rho \log z_{it} + \sigma_\epsilon \epsilon_{it+1}. \quad (7)$$

We further assume that firm innovation affects μ_{it} , which influences the long-run mean of productivity. This specification captures the idea that innovation creates growth options for generating future projects and it pushes forward a firm's technology frontier (e.g., Levine 2017). To maintain a long-run mean of productivity μ_{it} , firms incur research and development costs of innovation each period:

$$X(\mu_{it}, z_{it}) = \chi (\mu_{it} - \underline{\mu})^2 z_{it}, \quad (8)$$

where $\chi > 0$ is a scale parameter, and $\underline{\mu}$ is the lower bound attained when the firm chooses not to innovate (i.e., zero expenditures on R&D). The linear term z_{it} ensures that the cost of R&D scales up with the gains from innovation, a standard feature in models with endogenous productivity growth.

Firms choose their optimal innovation policy (i.e., the optimal μ_{it}) each period by maximizing the expected gains from innovation net of the R&D cost. The value of innovation arises from two sources: first, innovation improves a firm's productivity and thus its stand-alone value; and second, a more innovative firm demands a higher offer price when acquired and thus provides a better value at exit. We characterize firm-optimal innovation decisions in Section 2.4 after introducing the market for M&A.

2.3 The market for M&A

2.3.1 Merger technology. Firms also participate in an active M&A market and can potentially benefit from merging with other firms. Our model features

a random matching between firms. Specifically, each period, a firm meets with a potential merging partner, and upon the match, two firms examine the expected gains from merging together, which depend on the combined firm's productivity. We assume a Cobb-Douglas merger technology as in David (2021), that is, as an acquirer with productivity z_A merges with a target with productivity z_T , the combined firm's productivity $z_m = f(z_A, z_T)$ is determined as

$$f(z_A, z_T) = \gamma z_A^\alpha z_T^{1-\alpha} \quad (9)$$

with $\gamma > 0$, and $\alpha \in (0, 1)$. The parameter $\gamma > 0$ controls the scale of total value created in the merger, and the parameter α controls the relative share of contribution to the combined firm's productivity made by the acquirer and target. Intuitively, a higher value of $\gamma > 0$ implies more profitable mergers, and a higher value of α amplifies the role of the acquirer in the merger.

Equation (9) suggests that the combined firm's productivity depends on the role of the two merging firms in the M&A and therefore our model also characterizes the endogenous decision of "who acquires whom" in equilibrium.

2.3.2 Information structure in M&A. Information asymmetry is the key friction in our model, and it affects both merger terms (e.g., offer value, method of payment) and outcomes. We build this friction into our model by assuming asymmetric information on firm productivity z . Specifically, we assume that each firm i perfectly observes its own productivity at any point of time.³ Other firms, however, can only observe firm i 's productivity with a one-period lag. This is because any firm's past profits are observable, and Equation (6) suggests that productivity is linear in profits. As a result, public information of lagged profits constitutes the first channel of information flow, through which corporate outsiders learn about firm private information.

Mergers create the second channel of information flow in the model. As two firms meet and contemplate a potential merger, they spend more time learning about each other. The acquirer conducts due diligence before making an offer, and we assume that, with probability $1 - \omega \in (0, 1)$, the target's contemporaneous productivity is fully revealed, and thus the acquirer's information set becomes the same as the target's; with the complementary probability ω , the target's contemporaneous productivity remains private, and in this case, the acquirer has to form a rational expectation based on the target's lagged productivity. A larger value of ω therefore is associated with more substantial information frictions faced by the acquirer.

For the target firm, we assume that it always learns the acquirer's contemporaneous productivity through contact. Arguably, the main challenge of acquiring innovation results from the difficulty in evaluating target

³ We relax this assumption in Section 6.2, where we consider the possibility of firms not knowing their own contemporaneous productivity either.

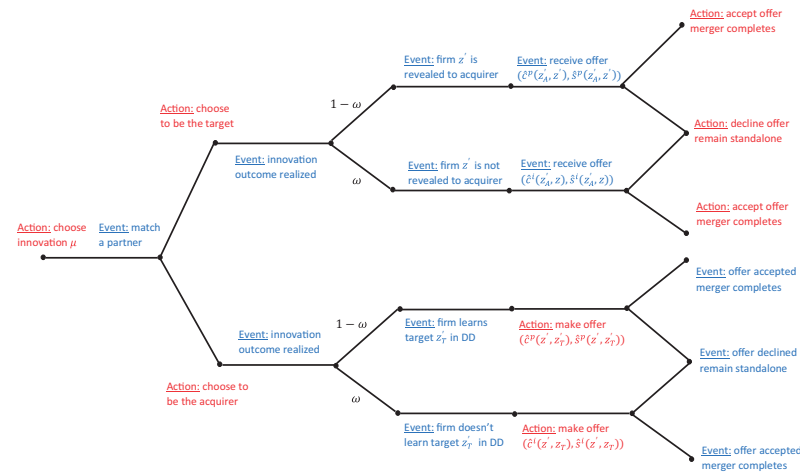


Figure 2
Model timeline

This figure illustrates the model timeline within a given period t . It describes the events and actions involving a firm in our model, as well as its information set. At the beginning of period t , the firm chooses its innovation level, and then randomly matches with a potential merger partner. The firm chooses its potential role in the merger, a decision that generates two branches: (1) If the firm acts as the target (the upper branch), it learns the contemporaneous productivity of itself and the acquirer (i.e., z' and z'_A) immediately after the innovation outcomes are realized. The acquirer then conducts due diligence (DD). With probability $1 - \omega$, the acquirer learns the firm's contemporaneous productivity, z' , and with probability ω , the acquirer learns nothing new. The firm then receives an offer from the acquirer and decides to accept or decline it. (2) If the firm acts as the acquirer (the lower branch), it learns its own contemporaneous productivity immediately after the innovation outcomes are realized. It, however, does not observe the target's new productivity. The firm then conducts due diligence to reveal the target's type with a probability of $1 - \omega$. Based on the firm's information set after due diligence, it makes an offer to the target, which may be accepted or declined. If an offer is accepted, then two matched firms merge together and the target firm exits; if an offer is declined, then both firms remain stand-alone.

innovation, and thus the target-side, rather than the acquirer-side, private information plays the pivotal role in our model. We discuss this assumption in more detail in Section 2.5 and analyze the effect of the acquirer-side information asymmetry in Internet Appendix B.1.

2.3.3 Timeline of a merger. We characterize a typical merger process in Figure 2. For a given period t , the figure illustrates the series of events a firm experiences and the sequence of actions it takes during this process.

Specifically, at the beginning of period t , the firm chooses its innovation level, μ , which affects its contemporaneous productivity, governed by Equation (7). The firm then randomly matches with a potential merger partner. Upon the match, the firm has to choose its potential role in the merger, either as the acquirer or as the target, and this decision generates two branches on the timeline.

If the firm acts as the target (the upper branch path), it learns the contemporaneous productivity of itself and that of the acquirer (i.e., z and z_A) immediately after the innovation outcomes are realized. The acquirer then conducts due diligence. With probability $1 - \omega$, the firm's contemporaneous productivity, z , is revealed to the acquirer, and with probability ω , the firm's contemporaneous productivity remains its private information. The firm then receives an offer from the acquirer that specifies the method of payment. The offer depends on the acquirer's information set after due diligence (i.e., whether the firm's contemporaneous productivity is revealed). The firm then decides whether to accept the offer and merge with the acquirer or to decline the offer and remain stand-alone.

If the firm acts as the acquirer (the lower branch path), it learns its own contemporaneous productivity immediately after the innovation outcomes are realized. It, however, does not observe the target's new productivity z_T right away. The firm then conducts due diligence that reveals the target's z_T with a probability of $1 - \omega$. Based on the firm's information set after due diligence, it makes an offer to the target, which may be accepted or declined. If an offer is accepted, then two matched firms merge together and the target firm exits; if an offer is declined, then both firms remain stand-alone.

We summarize the timeline of the merger process as follows:

1. Firms choose their innovation policy μ_i and incur the R&D cost $X(\mu_i, z)$.
2. Matches in the merger market are realized.
3. Two matched firms choose their roles as the acquirer or as the target.
4. Innovation outcomes are realized, and the contemporaneous productivity of the acquirer and the target are updated.
5. With probability $1 - \omega$, the target's contemporaneous productivity is revealed to the acquirer, and with probability ω , it remains the private information of the target.
6. The acquirer firm makes an offer to the target, specifying the method of payment.
7. The target decides whether to accept or decline the offer.

2.3.4 Decision to merge.

Acquiring firm: First, we consider the problem of a given acquiring firm with technology z_A that meets a target firm. There are two possible scenarios:

Scenario 1. If the target's contemporaneous productivity is not revealed, the acquirer only observes the target's last period productivity $\tilde{z}_T$ and would try to form a rational expectation of the target's contemporaneous productivity z_T based on its information set. We write the value function of the acquirer under this scenario as $V_A^i(z_A, \tilde{z}_T; \Theta)$, where the superscript i stands for *imperfect*

information. The maximization problem is set up as

$$V_A^i(z_A, \tilde{z}_T; \Theta) = \max_{c \geq 0, s \in [0, 1]} \left\{ \mathbb{E} \left[\mathbb{I}_T \left((1-s) \left(-c + \pi z_m - g(c, s, z_m; \Theta') \right) + \frac{1-\psi}{1+r} V(z_m; \Theta') \right) \right] + (1 - \mathbb{I}_T) \left(\pi z_A + \frac{1-\psi}{1+r} V(z_A; \Theta') \right) \middle| \tilde{z}_T, z_A; \Theta \right\} \quad (10)$$

$$\mathbb{I}_T = \hat{\mathbb{I}}_T(z_A, z_T, c, s; \Theta), \quad (11)$$

where Θ denotes the aggregate state of the economy, r is the risk-free discount rate, and ψ is the probability of exogenous exit.⁴ The acquirer chooses the optimal amount of cash c and the fraction of the combined firm's equity s to offer to the target. The first term in the above equation denotes the value to the acquirer if the offer is accepted ($\mathbb{I}_T = 1$), and the second term denotes the value to the acquirer if the offer is rejected.⁵ If the offer is accepted, the acquirer pays c to the target and gets $1-s$ fraction of the combined firm value. Empirically, equity payment is often accompanied by unfavorable market reactions to the acquirer's valuation, which constitute the costs of equity usage. Motivated by this empirical fact, we deduct a term $g(c, s, z_m; \Theta')$ from the combined firm's value:

$$g(c, s, z_m; \Theta') = \Lambda E(c, s, z_m; \Theta') z_m, \quad (12)$$

where the term $E(c, s, z_m; \Theta')$ is the equity ratio of the offer. With $\Lambda > 0$, the expected value loss due to equity usage is increasing in the equity ratio, and the costs of equity usage are zero if the offer is paid with 100% cash (i.e., $s=0$).

We choose to capture the costs of equity usage with the function $g(c, s, z_m; \Theta')$ for two reasons. First, our model characterizes the optimal offer choices made by an acquirer in face of the costs of equity usage, and therefore the magnitude of the costs is crucial to the model, while its microfoundation is of less importance. In other words, as long as the costs of equity usage are properly specified, the model solution and estimation are robust to the sources of such costs. Second, as we will discuss in Section 2.5, the specification of function $g(c, s, z_m; \Theta')$ is consistent with an extended model that incorporates the acquirer-side information asymmetry. Such an extension, however, does not provide additional insights to the model beyond the reduced-form specification

⁴ The aggregate state deterministically evolves to Θ' , and in a stationary equilibrium, $\Theta' = \Theta, \forall t$.

⁵ This is determined by the target's maximization problem, which we will describe below shortly. In equilibrium, rational expectations are fulfilled so that the acquirer correctly anticipates the target's decision rule, $\hat{\mathbb{I}}_T(z_A, z_T, c, s; \Theta)$.

of the function $g(c, s, z_m; \Theta')$, but would significantly complicate the model, as well as lead to the problem of equilibrium selection.

$\hat{c}^i(z_A, \tilde{z}_T; \Theta)$ and $\hat{s}^i(z_A, \tilde{z}_T; \Theta)$ denote the cash and equity components of the optimal offer, solved endogenously in the model, when the target's true productivity is not known. The hat notation is used to indicate that these functions are the optimal policy rules. In the rest of the paper, we will continue to use the hat notation to indicate optimal policy rules.

Scenario 2. If the target's contemporaneous productivity is fully revealed, then the acquirer perfectly observes z_T , and the acquirer's maximization problem is simpler:

$$V_A^p(z_A, z_T; \Theta) = \max_{c \geq 0, s \in [0, 1]} \left\{ \mathbb{I}_T \left((1-s) \left(-c + \pi z_m - g(c, s, z_m; \Theta') \right. \right. \right. \\ \left. \left. \left. + \frac{1-\psi}{1+r} V(z_m; \Theta') \right) \right) + (1-\mathbb{I}_T) \left(\pi z_A + \frac{1-\psi}{1+r} V(z_A; \Theta') \right) \right\} \quad (13)$$

$$\mathbb{I}_T = \hat{\mathbb{I}}_T(z_A, z_T, c, s; \Theta). \quad (14)$$

In this case, $\hat{c}^p(z_A, z_T; \Theta)$ and $\hat{s}^p(z_A, z_T; \Theta)$ denote the cash and equity components of the optimal offer. We use the superscript p for the value function and the optimal policy rules to represent this scenario under which the acquirer has *perfect information*.

Target firm: Now, consider the problem of a target firm with technology z_T that receives a merger offer of cash and equity, (c, s) , from an acquirer firm with technology z_A . The target firm decides whether to accept or reject the offer by comparing its continuation value and the offer price:

$$V_T(z_A, z_T, c, s; \Theta) \\ = \max_{\mathbb{I}_T \in \{0, 1\}} \left\{ \mathbb{I}_T \left[c + s \left(-c + \pi z_m - g(c, s, z_m; \Theta') + \frac{1-\psi}{1+r} V(z_m; \Theta') \right) \right] \right. \\ \left. + (1-\mathbb{I}_T) \left[\pi z_T + \frac{1-\psi}{1+r} V(z_T; \Theta') \right] \right\} \quad (15)$$

The first term represents the offer value perceived by the target and the second term is the target's continuation value if it rejects the offer. $\hat{\mathbb{I}}_T(z_A, z_T, c, s; \Theta)$ denotes the acceptance versus rejection decision that solves the above maximization problem.

2.4 Firms' innovation decision

Given the expected merger gains described above, we now can characterize firms' optimal innovation decisions. At the beginning of each period, firms

choose their innovation policy μ_i to maximize the expected firm value:

$$\begin{aligned}
 V(z; \Theta) = \max_{\mu_i \geq \underline{\mu}} & \left\{ \int \left[\mathbb{I}_{acq}(z, z_o) \left(\omega \mathbb{E} \left[V_A^i(z', z_o; \Theta) \mid z, \mu_i; \Theta \right] \right. \right. \right. \\
 & + (1 - \omega) \mathbb{E} \left[V_A^p(z', z_o; \Theta) \mid z, \mu_i; \Theta \right] \\
 & + (1 - \mathbb{I}_{acq}(z, z_o)) \left(\omega \mathbb{E} \left[V_T(z'_o, z', \hat{c}^i(z'_o, z), \hat{s}^i(z'_o, z); \Theta) \mid z, \mu_i; \Theta \right] \right. \\
 & + (1 - \omega) \mathbb{E} \left[V_T(z'_o, z', \hat{c}^p(z'_o, z'), \hat{s}^p(z'_o, z'); \Theta) \mid z, \mu_i; \Theta \right] \Big) dF_s(z_o) \\
 & \left. - X(\mu_i, z) \right\}, \tag{16}
 \end{aligned}$$

where $\mathbb{I}_{acq}(z, z_o)$ is an indicator function that takes the value one if the current firm acts as the acquirer in a given merger and zero otherwise,⁶ and $F_s(z)$ is the steady-state distribution of firms at the beginning of the period. z_o denotes the matched firm's productivity. Since matching is random, $z_o \sim F_s(z)$. Because innovation realizations occur after the acquirer and the target firms are chosen, new productivities, z' and z'_o , are random variables that depend on the productivities at the beginning of the period, z and z_o , and the associated innovation policies. The first term stands for the case when the firm is an acquirer and has imperfect information about the target's productivity, z'_o . The second term is for the case when the firm is an acquirer and has perfect information regarding the target. The third term stands for the case in which the firm is a target and receives an offer from an acquirer acting under imperfect information. The fourth term is the case in which the firm is once again a target, but the acquirer knows the firm's productivity z' . The final term is the cost of R&D necessary to maintain the innovation level μ_i . $\hat{\mu}_i(z; \Theta)$ denotes the optimal innovation policy function that solves the problem.

2.5 Model discussion

Like any economic model, our model makes a few important simplifying assumptions that allow us to focus on the mechanisms of interest, while maintaining tractability. In this section, we discuss their effects on our model solution and estimation.

First and foremost, our model focuses on characterizing the adverse selection risk that arises from the target-side private information. We believe that this focus is central to the problem we study in the paper. However, acquirers could possibly possess private information. In fact, the market timing hypothesis conjectures that acquirer-side private information induces overvalued acquirers to use equity as cheap currency in their takeover bids, which may trigger

⁶ The optimal decision to be the acquirer or the target maximizes ex ante surplus from the merger.

negative market reactions to equity usage. Since the payment method is crucial to the model solution, we capture the effect of acquirer-side private information on the method of payment by specifying a function $g(c, s, z_m; \Theta')$. This function characterizes the acquirer's costs of using equity in takeovers, and we let the data guide our estimation of the function parameters. In this sense, our estimation indeed "controls for" the effect of acquirer-side private information, and as long as the function $g(c, s, z_m; \Theta')$ is properly specified and estimated, our estimates are likely unbiased.

In Internet Appendix B.1, we present an extended model that explicitly incorporates the acquirer-side private information. We show that such a two-sided information asymmetry model gives rise to the well-known equilibrium multiplicity outcome and equilibrium selection becomes necessary. An appealing feature, however, is that all these equilibria predict the same discounting of equity by the target, which is consistent with our specification of $g(c, s, z_m; \Theta')$. This extension therefore provides a microfoundation for the function $g(c, s, z_m; \Theta')$ in our baseline model.

Second, mergers and acquisitions can be driven by various motives. The merger technology we specify in Equation (9) nests the Q-theory of M&A (Jovanovic and Rousseau 2002) as α approaches 1 and the asset complementarity theory (Rhodes-Kropf and Robinson 2008) as α approaches 0.5. Our model allows firm innovation to affect firm productivity and thus naturally embeds the incentive of acquiring innovation. Our model, however, does not feature imperfect competition. Using a model of M&A with oligopolistic competition in each industry, Cavenaile, Celik, and Tian (2021) show that only mergers between superstar firms in two superstar industries are "anticompetitive" in nature, whereas the mergers in remaining industry states are motivated by synergy gains instead. Cunningham, Ederer, and Ma (2021) also find that only about 6% of acquisitions in pharmaceuticals industry are killer acquisitions. Given the relatively small fraction of acquisitions with anticompetitive implications, we expect that leaving this motive out of our model would not significantly bias our estimates in our sample.

Third, our model makes a parsimonious assumption that information frictions arise from the latest shocks to the target's innovation outcome. This specification is consistent with the model setup that firm profitability is proportional to past innovation outcomes and thus observing historical profitability reveals innovation outcomes in the past. Following the innovation literature, we present motivating evidence (in Section 1) and estimate the model parameters (in Section 4) using innovation stock data in the paper. In Tables E1 to E3 in Internet Appendix E, we show that the empirical results that motivate our study and guide our estimation remain quantitatively similar if we use the innovation measured during the past 3 years instead, arguably a better proxy for firms' latest innovation activities.

Last, but not least, the model assumes that target firms observe their own contemporaneous innovation realization before accepting or declining the

M&A offer. One possible concern is whether this assumption is warranted and whether violation of this assumption would significantly bias our results. In Section 6.2, we extend the model to allow target firms to have imperfect information about their own innovation and evaluate the robustness of our results. We demonstrate that the inefficiencies caused by information frictions remain quite substantial even if target firms possess no private information in 30% of cases.

3. Model Solution

We solve the model numerically and characterize the model solution below. See Internet Appendix A for a formal definition of the unique stationary general equilibrium of the described economy and computational details. The model solution describes the acquirer's decision on offer price, the target's decision on whether to accept or decline the offer, and the firm's innovation policies. We also discuss how merger probability and deal completion rate are determined in the equilibrium.

3.1 Acquirer offer price

Acquirers need to determine the offers made to the targets. The offer price depends on whether or not the target's quality is revealed. When the target's quality is fully revealed, there is no information asymmetry between the two firms and the offer price is easy to determine. Our analyses below, therefore, focus on the situation when the target's quality is not fully revealed and the acquirer needs to form a rational expectation of the target's quality. Our results show that there exist cases in which the acquirer cannot perfectly screen the target's type and it makes an offer that is insufficient to attract the target of best quality. In this sense, information asymmetry leads to inefficiencies in the takeover market because the deals that have the potential of creating the largest value fail to consummate.

To elucidate the mechanism, we consider an example in which an acquirer with type z_A meets a target with preinnovation productivity $\tilde{z}_T$. The true productivity of the target, z_T , is not fully revealed, so the acquirer must conjecture based on the innovation policy of the target, $\hat{\mu}_i(\tilde{z}_T; \Theta)$ and the law of motion for productivity given by Equation (7). For simplicity, let's consider a discretized version of the model with only three possible types for the target: $z_T^h > z_T^m > z_T^l$ with $z_T^m = \tilde{z}_T$.⁷ The acquirer knows how each of the three types would react to an offer (c, s) . Figure 3 depicts the indifference curves of each target type, along which they are indifferent between accepting or rejecting the offer. Any offer above the curve would be accepted. In this example, an offer that is acceptable to z_T^h would be acceptable to both z_T^m and z_T^l , but the opposite

⁷ That is, a Tauchen discretization of $\mathbb{E}[z_T | \tilde{z}_T, \hat{\mu}_i(\tilde{z}_T; \Theta)]$ with three bins.

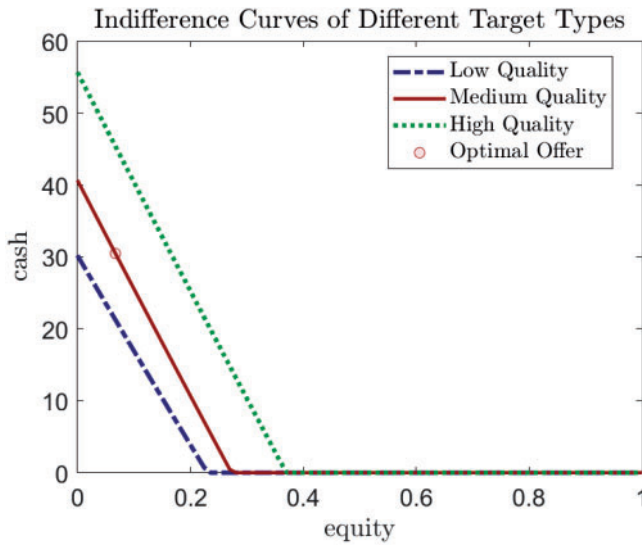


Figure 3
Target indifference curves

This figure illustrates an example in which an acquirer with type z_A meets a target with preinnovation productivity, $\bar{z}_T$. The true productivity of the target, z_T , is not public information, so the acquirer must form rational expectations based on the innovation policy of the target and the law of motion for productivity. For simplicity, we consider a discretized version with three possible types for the target: $z_T^h > z_T^m > z_T^l$ with $z_T^m = \bar{z}_T$. The acquirer knows how each of the three types would react to any offer, (c, s) . The figure depicts the indifference curves of each target type along which the acquirer is indifferent between accepting or rejecting the merger offer. Any offer above the curve would be accepted. In this example, an offer that is acceptable to z_T^h would be acceptable to both z_T^m and z_T^l , but the opposite is not true. Given this information, the acquirer chooses the amount of cash and equity in the offer, (c, s) , to maximize its expected value.

is not true. Given this information, the acquirer has to choose (c, s) to maximize its expected value.

Given the target's indifference curve that is perfectly anticipated by the acquirer, the left panel of Figure 4 presents a heat map that illustrates the acquirer's value by making different offers. Lighter color indicates higher value to the acquirer: the acquirer's value is low at the northeast corner, where the acquirer makes a very generous offer and overpays the target to a large extent. The acquirer's value is also low at the southwest corner, where the offer made by the acquirer is too low, and the deal fails to complete and thus generates no merger gains for the acquirer. The optimal offer price is set based on the acquirer's indifference curve that maximizes his expected gains, taking into account the target's response determined by the target indifference curve shown in Figure 3. In our example, the value function of the acquirer is maximized at the dot, and the offer contains both cash and equity. This offer is accepted by z_T^l and z_T^m but rejected by z_T^h .

The right panel of Figure 4 presents a heat map that illustrates the value function of a target with z_T^m . The value function is increasing in c and s above

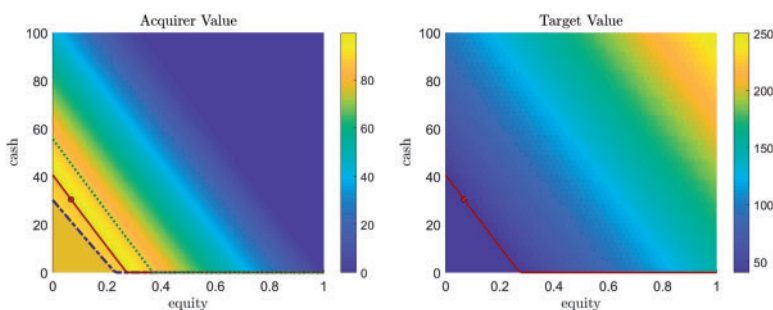


Figure 4
Acquirer and target value

The left panel of this figure displays the expected value of the acquirer as a heat map for the example in Figure 3. Given that there are only three types of targets, the optimal offer must lie on one of the target indifference curves, because any deviation from the indifference curve implies overpayment (the acquirer can pay less but still get the same set of targets to accept the offer). In this example, the value function of the acquirer is maximized at the dot, and the offer contains both cash and equity. This offer is accepted by the low and medium target types but rejected by the high type. The right panel of this figure displays the value function of the medium type target as a heat map. The value function is increasing in c and s above the indifference curve. However, it is flat below the indifference curve where its level is equal to the target's continuation value (reservation value). Therefore, the target type that marginally accepts the offer is only paid its outside option, whereas the low type is paid in excess of its outside option and collects information rents. This simple logic carries over to a discretization with more types, as well as a continuous z_T .

the indifference curve, because any offers with value above the indifference curve will be accepted and a more lucrative offer increases the target's value. However, the value function is flat below the indifference curve, because any offer below that will be rejected and then the value function is simply equal to the target's continuation value. Therefore, the target type that marginally accepts the offer, z_T^m , is only paid its outside option, whereas the low type z_T^l is paid in excess of its outside option and collects information rents. This logic carries over to cases with more target types or even a continuum of types.

3.2 Target response

We now present the target's response to takeover offers. If the target type is fully revealed through the acquirer's due diligence, the acquirer will make an offer that is always acceptable to the target as long as the surplus from merger is positive. Our analyses here therefore focus on the situation when the target type is not fully revealed. As we discussed above, the acquirer may make an offer that is appealing to only a subset of targets in this case. Let's assume that the acquirer has a productivity z_A and makes the optimal offer as discussed in the above section. We still assume that there are three types of targets, and $\tilde{z}_T$ denotes their preinnovation productivity.

Figure 5 plots the deal acceptance policy functions $\hat{\mathbb{I}}_T(z_A, z_T, \hat{c}^i(z_A, \tilde{z}_T), \hat{s}^i(z_A, \tilde{z}_T))$ of the three postinnovation types defined as before for multiple values of $\tilde{z}_T$, where a value of one indicates acceptance.

When $\tilde{z}_T$ is low, the acquirer makes an offer that would be accepted by all types. This happens because, as Equation (7) suggests, the effect of innovation

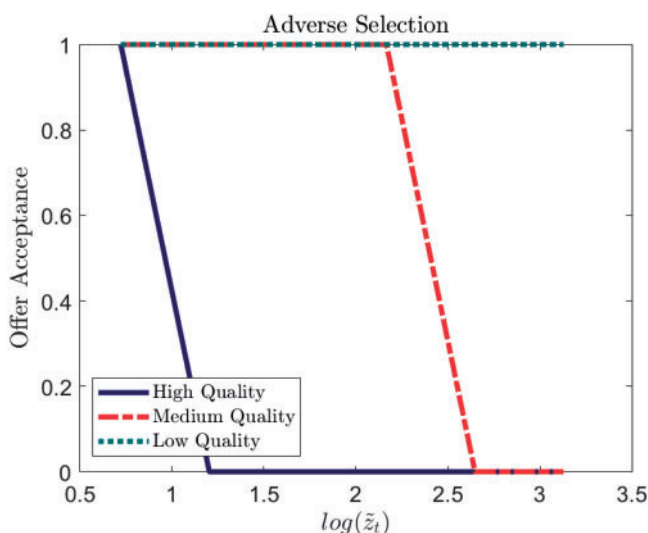


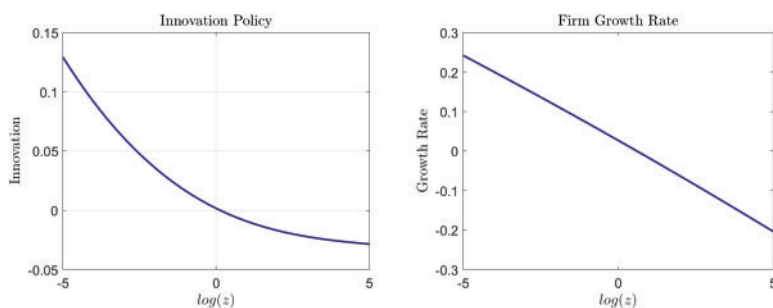
Figure 5
Offer acceptance: Adverse selection

This figure depicts the deal acceptance policy functions of the three postinnovation types defined in Figure 3 for multiple values of observed target innovativeness, $\tilde{z}_T$. Acceptance is represented by one. In this example, when $\tilde{z}_T$ is low, the acquirer makes an offer that is accepted by all types. This indicates that the expected value of the merger to the acquirer is so high that it does not mind overpaying the medium and low types. As $\tilde{z}_T$ increases, uncertainty becomes less tolerable. The adverse selection problem is aggravated, and the acquirer makes a more timid offer that the high type rejects but the medium and low types accept. For very high values of $\tilde{z}_T$, only firms with the lowest type accept the acquirer's offer, representing a market for lemons.

scales with the level of the lagged productivity $\tilde{z}_T$, and therefore when $\tilde{z}_T$ is low, uncertainty caused by the unobservable innovation is small in magnitude. In this case, the adverse selection problem induced by information frictions is less of a concern to the acquirer and it is able to make an offer that is generous enough to be accepted by all types of targets. In other words, the expected synergy is high enough to compensate the acquirer for the risk of overpaying the low and medium type of targets.

However, as $\tilde{z}_T$ increases, the adverse selection risk induced by information frictions intensifies and becomes more costly to the acquirer. The acquirer thus makes a more conservative offer, which is only accepted by the medium- and low-type targets. Even though the acquirer anticipates losing high-quality targets, it chooses the loss because the benefits of making a higher offer to attract high-quality targets are overwhelmed by the costs of significantly overpaying the lower-quality targets.

As $\tilde{z}_T$ becomes very high, the acquirer becomes even more conservative, because it understands that the costs of overpaying a low-quality target can be quite substantial in this case. That is, if it makes a very lucrative offer but eventually ends up acquiring a low-type target, the deal can come as a big disappointment. As a result, the acquirer chooses to make a very conservative

**Figure 6****Innovation policy and firm growth**

This figure depicts the optimal innovation policy function, and the implied expected firm growth rate for firms with different productivity levels z in the left and right panels, respectively. Consistent with the empirical findings in previous literature that reject Gibrat's law, smaller firms innovate more intensively and grow faster. The existence of an active M&A market creates an extra incentive for small firms to innovate more and grow faster, as becoming more productive makes them more valuable to potential acquirers. Large firms can choose a lower innovation intensity, as they have the option of improving their productivity by acquiring productive small firms rather than spending too much on in-house R&D.

offer that is accepted by the low-type target only. This is the classical market for lemons problem studied in Akerlof (1970). As adverse selection risk aggravates, it leads to the separation of low types from the medium and high types instead of the pooling results we observed above.

3.3 Innovation policy and firm growth

Figure 6 depicts the innovation policy $\hat{\mu}_i(z; \Theta)$, and the implied expected firm growth rate for firms with different productivity levels z in the left and right panels, respectively. Consistent with the empirical findings in previous literature that reject Gibrat's law (see Hall (1987), among others), smaller firms innovate more intensively and grow faster. The existence of an active M&A market creates an extra incentive for small firms to innovate, as becoming more productive makes them more valuable to potential acquirers. Large firms can choose a lower innovation intensity, as they have the option of improving their productivity by acquiring productive small firms rather than spending too much on in-house R&D.

3.4 Merger probability, deal completion, and offer composition

The above sections present model solutions as functions of state variables and illustrate the economic intuition behind them. To better link our model to the data, we examine in this section how the model-implied merger probability, deal completion rate, and the method of payment vary with the target firm's simulated stock of innovation in model equilibrium.⁸ To do so, we simulate the model and generate a cross-section of randomly matched pairs of potential

⁸ We construct the discounted stock of innovation for the simulated firms by using the same methodology from the empirical analysis. See Internet Appendix A for these details.

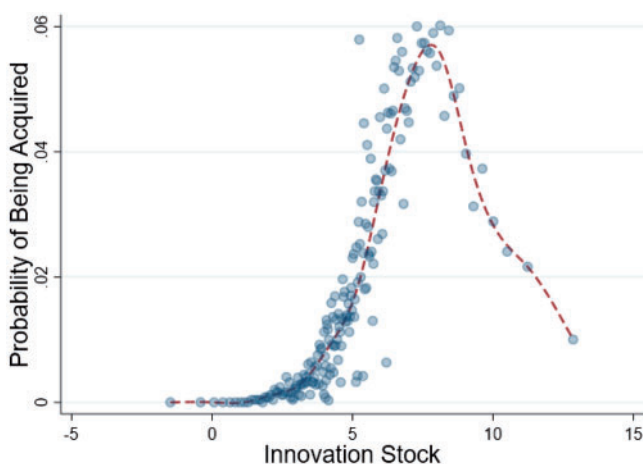


Figure 7
Probability of being acquired and firm innovativeness

This figure depicts the relationship between firm innovativeness (measured as the innovation stock of the firm) and the probability of being acquired. Scattered dots represent the simulated data from the estimated model, grouped into bins of innovation stock. The dashed line represents the fitted curve. When firm innovativeness is quite low, the benefit from M&A is too low for an acquirer to make an offer, and thus the probability of being acquired is zero. As firm innovativeness increases, so does its potential contribution as a target in M&A. Thus, the firm is more likely to receive offers, and the probability of being acquired increases. The probability of being acquired, however, starts decreasing as the firm becomes highly innovative due to an intensified adverse selection risk.

acquirers and targets.⁹ We track the deal outcomes and the method of payment in each deal. We then group the simulated data into bins based on target firms' stock of innovation and calculate the average merger probability, deal completion rate, and equity ratio within each bin. Figures 7 to 9 demonstrate our analyses.

Figure 7 depicts the model-implied relation between firm innovativeness and the probability of being acquired. When firm innovativeness is quite low, the benefit from M&A is small and thus the probability of being acquired is close to zero. As a firm becomes more innovative, its potential contribution as a target rises and thus the likelihood of being acquired gradually increases. The increased expected merger gain, however, is counteracted by the growing adverse selection risk, eventually leading to a hump-shaped relation between the takeover exposure and firm innovativeness. This happens because as a firm becomes more innovative, the realization of its ongoing innovation has a larger impact on its value and the expected synergy in M&A, leading to an intensified adverse selection risk. As a result, highly innovative firms are more difficult to be acquired due to the information friction.

Confirming our above discussion, Figure 8 illustrates a negative relation between the deal completion rate and target innovativeness. This happens because, facing an intensified adverse selection risk, acquirers tend to make

⁹ We use the estimated parameter values reported in Table 5 to solve and simulate the model.

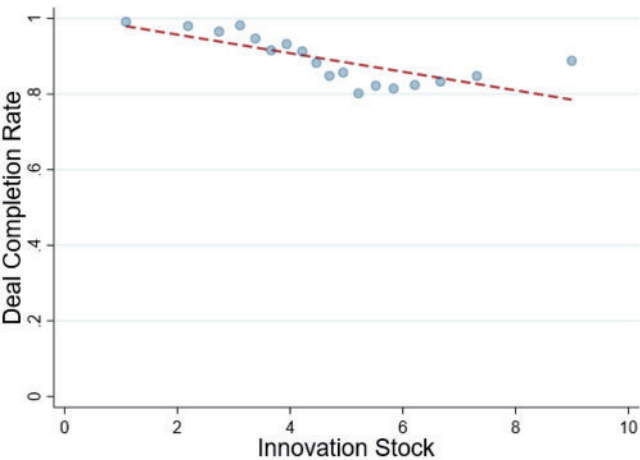


Figure 8
Deal completion and target innovativeness

This figure displays a negative relation between the likelihood of deal completion and target innovativeness (measured as the innovation stock of the target). Scattered dots represent the simulated data from the estimated model, grouped into bins of innovation stock. The dashed line represents the fitted curve. The deal completion rate drops as the target firm’s innovativeness increases, because acquirers are more reluctant to offer a high price in the face of adverse selection risk, which increases with the target’s firm innovativeness.

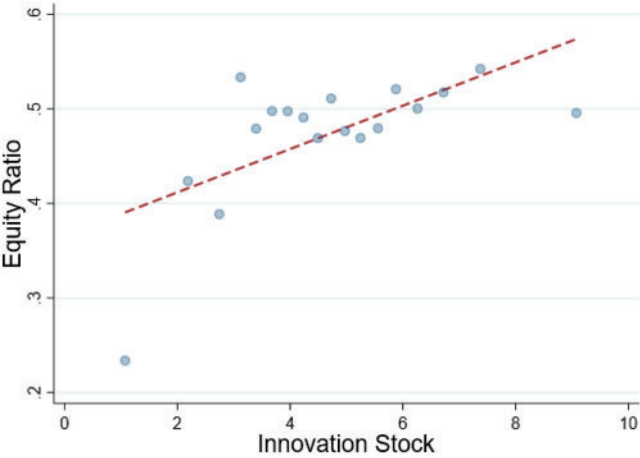


Figure 9
Offer composition and target innovativeness

This figure shows a positive relationship between the equity ratio in an offer and target innovativeness (measured as the innovation stock of the target). The scattered dots represent the simulated data from the estimated model, grouped into bins of innovation stock. The dashed line represents the fitted curve. Since acquirers cannot always perfectly observe the target’s true productivity under information asymmetry, they hedge the adverse selection risk by including equity in their offers. Adverse selection risk increases with target firm innovativeness, leading to an increased usage of equity as targets become more innovative.

more conservative offers to more innovative targets, and such offers are more likely to be turned down by high-quality targets.

Last, Figure 9 shows a positive relation between the equity ratio of the offer and target innovativeness. Since acquirers cannot always perfectly observe the target's true productivity under information asymmetry, they hedge the adverse selection risk by using equity in their offers. Adverse selection risk increases with the target firm innovation, and this leads to an increased usage of equity as the targets become more innovative.

4. Estimation and Identification

4.1 Identification

We start with parameters that are standard in the literature. The first three parameters are the subjective discount rate β , capital depreciation rate δ , and the production function concavity on capital κ . We set $\beta=0.9615$, consistent with a real interest rate of $r=0.04$, $\delta=0.069$ is taken from U.S. NIPA, and $\kappa=0.85$, consistent with an average markup of 18%.

We are left with nine parameters to be estimated: $\omega, \Lambda, \chi, \rho, \sigma, \mu, \psi, \gamma$, and α . We use 14 moments to identify the remaining 9 model parameters. Our identification strategy ensures that there is a unique parameter vector that makes the model fit the data as closely as possible. Since we estimate these parameters in one big SMM system, we essentially allow each moment to respond to all parameters in estimation. We will explain below which moments are the most informative in identifying each parameter.

1. **Information asymmetry:** The parameter ω determines the method of payment in our model. As the acquirer cannot always observe the target's contemporaneous productivity, it may prefer using equity for hedging purpose. Therefore, a higher value of ω increases the overall equity usage in the model. We thus use the average equity ratio of offers and the sensitivity of equity usage with respect to target innovativeness to help pin down ω . In particular, we repeat the regression outlined in Equation (2) using the model-simulated data, and seek to match the model-implied coefficient B_2 therein. Meanwhile, since equity usage is costly, acquirers are more reluctant to make a lucrative offer with equity payment, which reduces the likelihood of the offer being accepted. This is particularly true when acquirers bid for highly innovative targets where equity usage is crucial. So we also include the sensitivity of deal completion rate with respect to the target's stock of innovation as a targeted moment in SMM. Namely, we run the regression given in Equation (3) using the model-simulated data, and match the coefficient B_3 .
2. **Cost of equity usage:** Consistent with evidence, using equity is costly for acquirers due to the negative market reaction. The magnitude of this cost is governed by Λ . We estimate the value of Λ by replicating the empirical relation between equity usage and announcement returns. This

is done by targeting the implied value loss for acquirers obtained from the following regression:

$$AcqRet = A_4 + B_4 \cdot PrcOfStk + \varrho_4 \cdot X_{i,t} + \varepsilon_{i,t}. \quad (17)$$

Acquirer announcement return is regressed on the equity ratio, and the controls $X_{i,t}$ include the relative size of target, the size of acquirer, leverage, market-to-book ratio, return on assets, and cash to assets of the acquirer and target, as well as year and industry fixed effects. The implied value loss for acquirers is calculated as $|B_4 \cdot PrcOfStk|$.

3. **R&D cost:** The scale parameter of the R&D cost function, χ , determines the amount of resources firms spend on innovation. We target average R&D intensity defined as R&D expenditure over total assets to determine its value. This is a standard moment used in endogenous growth models.
4. **Acquirer's share of contribution in M&A:** The productivity of a combined firm is a Cobb-Douglas function of the productivities of the acquirer and the target. The parameter α determines the relative contribution of the acquiring firm. The relative size of the target firm to the acquirer helps identify α . As α increases, the acquiring firm contributes a larger fraction to the combined firm and thus the ratio of target-to-acquirer size shrinks accordingly.
5. **Evolution of firm productivity:** A firm's productivity evolves according to the AR(1) process given in Equation (7). There are three parameters to be estimated: persistence ρ , standard deviation of the innovation σ , and the lower bound of the drift term, $\underline{\mu}$. Since a firm's productivity determines its market value, the first two parameters are disciplined by targeting the estimated autocorrelation of log market value and its coefficient of variation. To pin down ρ , we run the following regression:

$$\ln(ME)_{i,t} = A_5 + B_5 \ln(ME)_{i,t-1} + \varrho_5 \cdot X_{i,t} + \varepsilon_{i,t}, \quad (18)$$

where $\ln(ME)$ is the natural logarithm of market equity and the regression controls for year fixed effects. We repeat the same regression using model-simulated firm data, and use the coefficient B_5 in Equation (18) as a target. To pin down σ , we run another regression given by

$$\ln(ME)_{i,t} = A_6 + \varrho_6 \cdot X_{i,t} + \varepsilon_{i,t}, \quad (19)$$

where the controls $X_{i,t}$ include leverage, market-to-book ratio, return on assets, cash to assets, tangibility, age, and squared age, as well as year and industry fixed effects. We obtain the residuals from this regression, which are purged from the impact of observables, and calculate its standard deviation and divide it by the sample mean of $\ln(ME)_{i,t}$ to obtain a coefficient of variation for log market value. We repeat the same regression using model-generated data and seek to match the estimated

coefficient of variation. This leaves us with the final parameter $\underline{\mu}$. An increase in $\underline{\mu}$ increases the growth rate of all firms since it lowers the cost of innovation $X(\mu_{it}, z_{it})$ given in Equation (8) across the board for any productivity drift μ_{it} . Hence, its value is pinned down by targeting the average firm growth rate.

It is worth noting that both ρ and σ are also important determinants of information frictions in our model. It is straightforward to see that information asymmetry and the adverse selection risk it induces are both increasing in σ . The effect of ρ , however, is more nuanced. On the one hand, a high value of ρ implies a more predictable AR(1) process of the target productivity. On the other hand, interestingly, a more persistent AR(1) process also implies that any shocks to the target's productivity would have a larger impact on target valuation, and as a consequence, the adverse selection risk induced by a given level of information asymmetry is actually increasing in ρ . To see the intuition, imagine an acquirer who cannot observe the realization of the target's ongoing innovation and its effect on productivity. This acquirer will be more concerned with the adverse selection risk if the target's productivity is more persistent, because in this case, a negative shock will have a long-lasting effect on the target's productivity and thus hurt the target valuation to a larger extent. As a result, moments related to information asymmetry provide additional identification power for ρ and σ as well.

6. **Exogenous exit rate:** In the model, firms can exit due to endogenous merger decisions, or exogenous shocks. The parameter ψ denotes the latter's probability. In a stationary equilibrium, the entry rate of new firms equals the exit rate. We pin ψ down by targeting the average entry rate of firms in our sample.
7. **Synergy:** γ is the scale parameter of the Cobb-Douglas aggregator that determines the productivity of a merged firm. $\gamma=2$ would imply that a merger between two identical firms would yield no productivity gains. Values above 2 capture the gains in productivity due to synergy. The value of this parameter is pinned down by targeting the average realized gain in market value (i.e., the combined firm's announcement period abnormal return) and the average merger probability, which are both increasing in γ .

To further demonstrate the identification of model parameters, we report the model-implied sensitivity of moments with respect to parameter values in Table E9 in the Internet Appendix (i.e., the Jacobian matrix). We place the parameters on columns and model-implied moments on rows. To make a clear presentation of the identification, we reorder the parameters and moments in this table so that the parameters are lined up with the main identifying moments around the diagonal (the order of parameters and moments are therefore slightly different from that reported in Table 5). The table reports how model-implied

Table 5
Baseline model parameters and target moments

A. Parameter estimates

Parameter	Description	Value	SE
ω	Information asymmetry	0.697	0.010
Λ	Market reaction to equity usage	0.051	0.001
χ	Innovation cost	58.194	0.727
ρ	Productivity persistence	0.863	0.003
σ	Innovation SD	0.374	0.002
$\underline{\mu}$	Lower bound of productivity drift	-0.032	0.001
ψ	Exogenous exit rate	0.024	0.001
γ	Merger technology, scale	3.636	0.042
α	Merger technology, acquirer share	0.644	0.007

B. Moments

Target moments	Model	Data	SE
Average equity ratio	49.63%	47.17%	1.87%
Loading of equity ratio on <i>innovstock</i> ($B_2 \times 100$ in Equation (2))	2.282	2.432	0.282
Loading of merger prob. on <i>innovstock</i> ($B_1 \times 100$ in Equation (1))	0.865	0.602	0.011
Loading of merger prob. on <i>innovstock</i> ² ($C_1 \times 100$ in Equation (1))	-0.020	-0.068	0.002
Average merger probability	1.65%	1.91%	0.06%
Loading of deal completion on <i>innovstock</i> ($B_3 \times 100$ in Equation (3))	-2.358	-2.953	0.233
Average realized gain	5.32%	3.16%	0.54%
Average value loss	0.68%	1.09%	0.04%
Average R&D intensity	5.84%	8.50%	1.77%
Autocorr. of $\ln(ME)$ (B_5 in Equation (18))	0.965	0.944	0.001
Coefficient of variation of $\ln(ME)$ from Equation (19)	0.210	0.364	0.004
Average firm growth rate	8.62%	5.21%	0.10%
Firm entry rate	4.09%	4.57%	0.07%
Average relative size (target/acquirer)	0.316	0.337	0.014

This table reports estimation results obtained from the simulated method of moments (SMM), which chooses model parameters by matching the moments from a simulated panel of firms to the corresponding moments from the data. Panel A reports the estimated parameters. Panel B reports the simulated and actual moments. See Section 4 for the definition of moments.

moments change as we perturb the model parameters. We want to emphasize two caveats for this table. First, even though we explain how some moments are more informative in identifying certain parameters, the SMM estimator extracts information from all moments in order to infer parameter values in the estimation process. The mapping between the moments and parameters are clearly not one to one. Second, we focus on explaining the economic intuition behind the links between some moments and certain parameters. Even though these moments often carry a large magnitude in the Jacobian matrix, which is consistent with our discussion of identification above, there are occasionally other moments that have an even larger magnitude. This is not a challenge to our discussion of identification, but simply implies that other moments may carry important information regarding the parameter of interest too, which necessitates the use of the SMM estimator.

4.2 Estimation results

We solve the model numerically and estimate it using SMM. Panel A of Table 5 summarizes the estimated parameter values. The parameter ω , which determines the extent of information asymmetry, is estimated to be 0.697. This estimate implies an only 30% of chance that target private information is fully revealed to the acquirer during the merger process. Information asymmetry, therefore, is substantial, and it creates significant barriers in the market for corporate control and hurts efficient asset reallocation in the economy.

Acquirer announcement returns are negatively correlated with equity usage in the data, and the estimated $\Lambda = 0.051$ helps the model fit this data feature fairly well. Though our model does not endogenize the mechanism of this market reaction to the method of payment, a positive Λ captures the costs of using equity payment in takeover offers. It leads to a critical tradeoff in the method of payment: acquirers prefer using equity to hedge adverse selection risk, but equity payment is accepted at a discounted price due to the negative market reaction to the announcement of equity bids.

The parameter of the R&D cost function, $\chi = 58.2$ delivers an average R&D intensity of 5.84%. The persistence of productivity $\rho = 0.863$ is quite high. This delivers a persistence of log firm value of 0.965; very close to the empirical value of 0.944 observed in the data. It suggests that the productivity gains from successful innovation are long-lasting. The standard deviation of innovation $\sigma = 0.374$ is large compared to the mean of $\log(z) = -1.29$. It indicates that there is a non-negligible amount of information asymmetry regarding the innovation of a firm conducted in recent periods.

The estimated probability of exogenous exit ψ is 0.024. Combined with the endogenous exit due to M&A at 0.0165, this delivers a firm entry rate of 4.09%, which is close to the value of 4.57% in our sample, and the overall Compustat entry rate of 5% found in Acemoglu, Akcigit, and Celik (Forthcoming).

We estimate γ , the scale factor in merger technology, to be 3.636. A benchmark value of $\gamma = 2$ implies that there is no synergy gain when two identical firms merge together. Our estimate of this parameter indicates that, on average, observed mergers create significant value for the merging pairs.

The parameter α that controls the acquirer's contribution to the combined firm's postmerger productivity is estimated to be 0.644. It implies that the combined firm's productivity is influenced more by the acquirer than by the target. This finding is consistent with the fact that, on average, the merged firm's value depends more on the acquirer's premerger value than that of the target.

All parameters are estimated with small standard errors, suggesting that these point estimates are relatively accurate. Two factors contribute to accuracy. First, most data moments are measured with high precision. Second, the model-implied moments are sensitive to parameter values, a feature that indicates strong identification power of the model.

Panel B of Table 5 reports the model fit by comparing the targeted data moments with their model counterparts.¹⁰ The model manages to match many data features fairly well. For example, in both the model and the data, the equity usage is close to 50% on average, and it increases across different deals with the target firm's innovativeness.

The relation between a firm's innovativeness and its takeover exposure is hump shaped, with the firms with an intermediate level of innovativeness being acquired most often in the data. When a firm's takeover exposure is regressed on the linear and quadratic terms of its innovativeness, the model produces a positive loading on the linear term and a negative loading on the quadratic term, consistent with the data pattern.

The average merger probability is low, and only less than 2% of firms get acquired every year, both in the model and in the data. This value is slightly lower than the average merger rate for the universe of Compustat firms, because firms in our sample are larger on average. Also, the likelihood of deal completion declines with target firm innovativeness, and the model does a good job in capturing this negative association accurately.

The model fit is slightly worse in some other dimensions. For example, the likelihood of being acquired declines slightly faster with firm innovation in the data than that in the model, especially for very large and innovative firms. This discrepancy happens because the model leaves out other factors in the real world that may prevent large, innovative firms from being acquired. For instance, financing costs often make it infeasible for smaller firms to acquire such large and innovative firms in the M&A market. The model also tends to slightly overstate the average realized gain and understate the average value loss in M&A. This is because our model does not feature value destroying mergers due to other frictions (e.g., agency costs). The empirical relation between announcement returns and equity usage allows the model to capture only part of the negative market reaction to merger announcements, and therefore it is not surprising that the model-implied realized gain (value loss) is higher (lower) than that in the data. The model does not fit as tightly in R&D intensity and firm growth as other moments. This happens because there exist some very large, innovative firms in the data, and with Gaussian distributed shocks to productivity, the model simulated distribution cannot perfectly cover these superstar firms. These firms inflate the average R&D intensity and lower the average firm growth rate in the data (i.e., large firms on average have a lower growth rate than smaller firms due to their large base to start with).

Overall, that the model is able to match the key features of firm innovation and takeover exposure in the data well lends support to the central mechanism

¹⁰ To make the coefficients of model-simulated regressions comparable, we rescale the target's innovation stock in the model such that it has the same mean and standard deviation as the target innovation stock in the data as measured by patent citations.

we hope to capture in this paper, namely, the information frictions in acquiring innovation.

5. Model Implications

5.1 Reducing information frictions

The main goal of our study is to quantify the effect of information frictions on acquisitions of innovation. In our model, conditional on a match, the acquirer cannot observe the true type of the target with probability ω . Intuitively, a lower value of ω can be interpreted as the acquirer being more capable of evaluating the true value of the target's contemporaneous innovation. Conversely, if innovation in the economy is more opaque, it would imply a higher value of ω . If $\omega = 1$, acquirers have no chance to learn the targets' contemporaneous innovation. In this section, we evaluate the effect of information frictions on corporate innovation and firm productivity growth through the channel of an active M&A market.

To do so, we carry out two counterfactual analyses using the estimated model. In the first counterfactual analysis, we fix firms' innovation policy and their cross-sectional distribution as in the baseline model equilibrium. We set $\omega = 0$ and allow the firms to reoptimize their decisions in M&A. Specifically, acquirers choose new optimal offers with perfect information about target innovation, and targets update their acceptance/rejection decisions accordingly. This exercise captures the short-term effect of eliminating information frictions, because we hold firm innovation policy and distribution unchanged. This analysis, therefore, shows the direct effect of information frictions on the M&A market.

In the second counterfactual analysis, we still set $\omega = 0$, but we now allow the firm distribution to evolve endogenously and let firms reoptimize their innovation policy. This exercise captures the long-term effect of eliminating information frictions, and it measures the changes in efficiencies as we move from one equilibrium to another.

To facilitate our analyses, we decompose a firm's market value into two components: its stand-alone value derived from shutting down its opportunity to participate in the M&A market during its lifetime, and its capitalized expected gains from participating in the M&A market.

Table 6 reports the results. The top four rows present a decomposition of firm market value into its stand-alone value and the capitalized expected gains from the M&A market, and the bottom five rows show the aggregate-level results. Column 1 reports the baseline model results. For an average firm in our estimated model, its stand-alone value accounts for about 94% of the market value, and its capitalized gains from future M&A represents the remaining 6%. An average firm has an R&D intensity around 6% every year, and expands at a rate of 8.6%. On average, 1.65% of firms get acquired each year.

In column 2, we evaluate the short-term effect of information frictions as described above. Since we fixed the firms' innovation policy and the

Table 6
Eliminating information frictions

	Baseline	Short-term effect ($\omega=0$) baseline $F_S(z)$ and $\hat{\mu}_i(z; \Theta)$	Long-term effect ($\omega=0$) new $F_S(z)$ and $\hat{\mu}_i(z; \Theta)$
Market value	11.048	11.577	11.348
Stand-alone value	10.392	10.392	10.304
Capitalized expected gain from M&A	0.657	1.185	1.045
Capitalized expected gain from M&A/market value	5.94%	10.24%	9.20%
Avg. R&D intensity	5.84%	5.84%	6.43%
Aggregate output	4.460	4.460	4.578
Avg. merger probability	1.65%	2.30%	1.95%
consumption	2.002	2.009	2.063
Avg. firm growth rate	8.62%	8.41%	8.87%

This table reports model implications in the baseline model and in a counterfactual economy in which information frictions are eliminated by setting $\omega=0$. Column 1 shows the baseline results. In column 2, we evaluate the short-term effect of information frictions, keeping firm innovation policy and the cross-sectional distribution as they are in the baseline equilibrium. Column 3 reports the long-term effect in which we allow firm distribution to evolve endogenously and let firms reoptimize their innovation policies. Market value is the model-implied value of the firm. We decompose the market value into two components: a stand-alone value derived by shutting down a firm's opportunity to participate in the M&A market during its lifetime, and an option value derived from the capitalized gains from all future M&A deals. R&D intensity is measured as the total expenses firms invest in innovation divided by firm assets; output is the aggregate production in the economy; avg. merger probability is the total number of mergers divided by the total number of firms in the economy; firm growth rate is the average firm sale growth.

cross-sectional distribution as they are in the baseline equilibrium, firm stand-alone value remains the same. The capitalized expected gains from M&A, however, increase significantly from 0.657 to 1.185, rising from 6% of the market value to more than 10% of firm value. Without information frictions, M&As are more likely to occur in the economy, and the merger rate climbs from 1.65% to 2.30%, representing a 40% increase. Since this counterfactual analysis holds firm innovation policies and the cross-sectional firm distribution unchanged, aggregate implications remain quite similar to those in the baseline.

We then carry out the second counterfactual analysis by allowing firm distribution to evolve and let the firms to reoptimize their innovation policies. In other words, we compute the full long-run general equilibrium effect. Column 3 confirms that both firm innovation activity (i.e., average R&D intensity) and total production (i.e., aggregate output) increase. Specifically, in the absence of information frictions, firms increase their innovation inputs by about 10%, and the aggregate output rises by 2.65%. Firm growth rate is also boosted by 25 basis points (bps) on average.

It is also interesting to note that the long-term effect of information frictions on the M&A market, reported in column 3 of Table 6, is slightly smaller than the short-term effect reported in column 2. For example, the capitalized gains from future M&A equals 1.045 in the long run, which is lower than 1.185 in the short run, and the merger probability is 1.95% in the long run versus 2.3% in the short run. This happens because in the long run, the firm distribution adjusts and many firms of medium innovativeness have been acquired, leaving fewer profitable M&A opportunities in the economy (compared with the short-run

effect where the firm distribution is held constant). In general, we can view the long-run effect as a new equilibrium in which information frictions are absent and the short-run effect as part of the trajectory moving from the old equilibrium to a new equilibrium. Eliminating information frictions, therefore, increases firms' capitalized expected gains from the M&A market by 59% in the long run ($59\% = \frac{1.045 - 0.657}{0.657}$).

Our analyses thus suggest that the inefficiency brought about by information asymmetry is quite sizeable, representing about 3.5% loss in a firm's market value (i.e., $0.035 = \frac{1.045 - 0.657}{11.048}$). A natural question is how this inefficiency compares with the costs a firm is willing to pay to reduce information asymmetry. For example, is it profitable for an acquiring firm to invest costly efforts in the due diligence process to mitigate the adverse selection risk? To answer this question, we note that the 3.5% value loss arises from information frictions embedded in all M&A deals the firm is expected to be involved in its lifetime. In other words, to recover this value loss, the firm needs to invest in due diligence in not only the forthcoming deal but also all future deals. The capitalized costs of such investment can be substantial, close to or even surpassing the 3.5% efficiency loss.

We can also look at the aggregate macroeconomic implications of shutting down information frictions. In the long run, the absence of information frictions increases aggregate output by 2.65% due to the permanent increase in average firm productivity. The increased efficiency also benefits the representative consumer, and we find that consumption-equivalent welfare increases by 3.02%, implying a substantial social gain.¹¹

5.2 Alternative benchmarks

The benchmark we use in the counterfactual analyses above assume that all information frictions are eliminated. In reality, fully eliminating information asymmetry is probably not achievable. To evaluate how our model implications change when only part of the information frictions can be removed, we experiment with a few alternative benchmarks in this section. We assume that practical policies can only mitigate a fraction (25%, 50%, and 75%) of the information frictions (i.e., set ω to 75%, 50%, and 25% of its estimated value). We present the model implications in these alternative benchmarks in Table 7. As expected, efficiency improvement relative to the alternative benchmarks is smaller. For example, eliminating 50% of information frictions (i.e., moving from baseline to $\omega = 50\% \times \omega^*$) increases firms' capitalized expected gains from M&A by 27% in the long run ($27\% = \frac{0.832}{0.657} - 1$), compared with 59% at the benchmark, where all information frictions are eliminated.

Research has shown that, in the course of conducting due diligence in M&A transactions, lawyers and auditors tend to focus their efforts mainly on the

¹¹ See Internet Appendix A.1 for the details of how welfare figures are calculated in general equilibrium.

Table 7
Alternative benchmarks

	Baseline	$\omega = 75\% \times \omega^*$	$\omega = 50\% \times \omega^*$	$\omega = 25\% \times \omega^*$	$\omega = 0$
Market value	11.048	11.119	11.181	11.274	11.348
Stand-alone value	10.392	10.356	10.348	10.317	10.304
Capitalized expected gain from M&A	0.657	0.763	0.832	0.957	1.045
Capitalized expected gain from M&A/market value	5.94%	6.86%	7.45%	8.49%	9.20%
Avg. R&D intensity	5.84%	5.93%	6.03%	6.33%	6.43%
Aggregate output	4.460	4.488	4.492	4.550	4.578
Avg. merger probability	1.65%	1.75%	1.77%	1.88%	1.95%
Consumption	2.002	2.018	2.023	2.048	2.063
Avg. firm growth rate	8.62%	8.70%	8.73%	8.81%	8.87%

This table reports model implications in the baseline model and in counterfactual economies in which information frictions are reduced to 75%, 50%, 25%, and 0% of the baseline value, respectively. Column 1 shows the baseline results. All columns report the long-term effect in which we allow firm distribution to evolve endogenously and let firms reoptimize their innovation policies. Table 6 defines the variables.

ownership of the company, equity, company debt, as well as the general physical structure of the company. Since most intangible assets are not recognized in financial statements and current accounting rules do not require firms to report separate measures for intangibles, corporate bodies and regulatory authorities inevitably pay little or even no attention to intellectual property (IP) rights. As part of the model’s implications on concrete policies, we believe that policies recommending more due diligence on intangible assets and policies that require improved disclosure of intangible assets can help reduce information asymmetry in acquisitions of innovative firms. In fact, due diligence on IP and intangible assets is indeed receiving growing attention among practitioners and corporate managers involved in the M&A transactions in recent years, even though academic research seems to lag behind on this front (see, e.g., De Schrijver and Demeyer 2018; Okojie 2018; Negi 2020; Pohl and Haughey 2021).

5.3 Equity usage

In the face of the adverse selection risk, acquirers can use equity as payment to bring target shareholders on board and reduce the overpayment risk. Equity usage, therefore, helps mitigate the effect of information frictions and facilitates deal completion. In this section, we carry out a counterfactual analysis in which we force acquirers to use only cash as payment and examine how this choice changes our results.

Table 8 presents the findings. We notice that shutting down equity usage reduces firms’ capitalized gains from M&A by almost 40% in both the short run and long run (i.e., $\frac{0.404-0.657}{0.657}$), which drops from about 6% of firm value in the baseline model to less than 4% in this counterfactual model with no equity usage. Since acquirers are not allowed to use equity to hedge the adverse selection risk, they become even more conservative in making offers to highly innovative targets, making acquisitions of innovation more challenging.

Table 8
No equity usage

	Baseline	Short-term effect baseline $F_S(z)$ and $\hat{\mu}_i(z; \Theta)$	Long-term effect new $F_S(z)$ and $\hat{\mu}_i(z; \Theta)$
Market value	11.048	10.796	10.803
Stand-alone value	10.392	10.392	10.394
Capitalized expected gain from M&A	0.657	0.404	0.408
Capitalized expected gain from M&A/market value	5.94%	3.74%	3.78%
Avg. R&D intensity	5.84%	5.84%	5.19%
Aggregate output	4.460	4.460	4.323
Avg. merger probability	1.65%	1.47%	1.56%
Consumption	2.002	2.009	1.954
Avg. firm growth rate	8.62%	8.67%	8.49%

This table reports model implications in the baseline model and in a counterfactual economy in which equity usage is prohibited. Column 1 shows the baseline results. In column 2, we evaluate the short-term effect of shutting down equity payment, keeping firm innovation policy and the cross-sectional distribution as they are in the baseline equilibrium. Column 3 reports the long-term effect in which we allow firm distribution to evolve endogenously and let firms reoptimize their innovation policies. Table 6 defines the variables.

A less efficient M&A market also has broader impacts on the macroeconomy, and we observe that aggregate output shrinks by 3.1% of its baseline value, which implies a considerable loss in social welfare at 2.40%. Innovation and business dynamism are also adversely affected: Average R&D intensity falls by 11.15% of its base value, and the average firm growth rate falls by 13 bps.

Overall, this counterfactual analysis sheds light on the value of equity usage in mitigating the negative effect of information frictions in the M&A market.

6. Subsample Estimation and Robustness Checks

6.1 Subsample estimation

Given the pivotal role of information frictions, we perform a subsample estimation in this section to further explore the cross-sectional variation of our results. We partition firms into two subsamples based on their ex ante measure of information asymmetry and then estimate our model in the two subsamples. The goal is to examine whether our estimated parameters can pick up the different information environments faced by firms in different subsamples.

To construct the ex ante measure of information asymmetry, we obtain Earnings per Share (EPS) forecasts at the firm-analyst level from the Institutional Brokers Estimates System (I/B/E/S) database. We also obtain the realized values of firm EPS from I/B/E/S. For each firm-year observation, following Terry (2017), we focus on the analysts' EPS forecasts in a two-quarter horizon, that is, from 91 to 180 days before the data release. We first measure each analyst's forecast error for a given firm-year as the difference between each analyst's forecast and the realized value of firm EPS, normalized by the absolute value of realized EPS. Then, for each firm-year observation, we compute the root-mean-square error (RMSE) of the forecast errors. We use the RMSE of analysts' forecast errors as the ex ante measure of information asymmetry. For each industry in our sample, we calculate the mean of the

Table 9
Parameter estimates for high and low analyst forecast error subsamples

ω	Λ	χ	ρ	σ	$\underline{\mu}$	ψ	γ	α
<i>Low-RMSE analyst forecast error subsample</i>								
0.6543 (0.0104)	0.0546 (0.0009)	60.060 (0.5496)	0.8698 (0.0019)	0.3615 (0.0057)	-0.0327 (0.0011)	0.0242 (0.0007)	3.6243 (0.0043)	0.6359 (0.0027)
<i>High-RMSE analyst forecast error subsample</i>								
0.7494 (0.0158)	0.0571 (0.0053)	45.820 (4.0319)	0.8589 (0.0071)	0.3886 (0.0024)	-0.0324 (0.0015)	0.0253 (0.0006)	3.5850 (0.0387)	0.6442 (0.0059)

This table reports the estimated parameters for subsamples partitioned based on information asymmetry, and the standard errors of parameter estimates are reported in parentheses. To construct the ex ante measure of information asymmetry, we obtain Earnings per Share (EPS) forecasts at the firm-analyst level from the Institutional Brokers Estimates System (I/B/E/S) database. We also obtain the realized values of firm EPS from I/B/E/S. We first measure each analyst's forecast error for a given firm-year as the difference between each analyst's forecast and the realized value of firm EPS normalized by the absolute value of realized EPS. Then, for each firm-year observation, we compute the root-mean-square error (RMSE) of the forecast errors. For each industry in our sample, we calculate the mean of the RMSE of analysts' forecast errors. We rank industries according to this mean and split the full sample into high- and low-RMSE subsamples.

RMSE of analysts' forecast errors. We rank industries according to this mean and split the full sample into high- and low-RMSE subsamples.

As a first-pass check of these two subsamples, we notice that firms in the high-RMSE subsample use more equity in their M&A transactions compared with those in the low-RMSE subsample (51% vs. 42%). Deal completion rate is also much more sensitive to the level of target innovation in the high-RMSE subsample, compared with the low-RMSE subsample (with a slope of -0.04 vs. -0.03). Firms in the high-RMSE subsample conduct more R&D than those in the low-RMSE subsample (with an R&D intensity of 11% vs. 6%). These patterns are statistically significant and consistent with our expectation of RMSE being an ex ante measure of information asymmetry associated with firm innovation.

We then estimate our model in the two subsamples and report the parameter estimates in Table 9. First and foremost, the estimated ω is 0.75 for the high-RMSE subsample and 0.65 for the low-RMSE subsample. The difference is statistically significant and economically sizeable. Since firms in the high-RMSE subsample invest more on R&D, their estimated costs of innovation χ is also significantly lower than their counterparts in the low-RMSE subsample. Other parameter estimates remain quantitatively close in the two subsamples, implying that the differences in these two subsamples are mainly driven by information asymmetry and the associated innovation.

Lastly, we quantify the effect of information asymmetry in these two subsamples using the estimated model parameters. Table 10 presents the results. As expected, eliminating information asymmetry (i.e., set $\omega=0$) increases firms' capitalized expected gains from M&A by a much larger margin for the high-RMSE subsample, compared with the low-RMSE subsample ($\frac{1.049-0.633}{0.633} = 66\%$ vs. $\frac{0.953-0.654}{0.654} = 46\%$).

Table 10
Eliminating information frictions for high and low analyst forecast error subsamples

	Low-RMSE subsample		High-RMSE subsample	
	Baseline	$\omega=0$ new $F_S(z)$ and $\hat{\mu}_I(z; \Theta)$	Baseline	$\omega=0$ new $F_S(z)$ and $\hat{\mu}_I(z; \Theta)$
Market value	10.831	11.084	11.038	11.377
Stand-alone value	10.177	10.131	10.405	10.329
Capitalized expected gain from M&A	0.654	0.953	0.633	1.049
Capitalized expected gain from M&A/market value	6.04%	8.60%	5.73%	9.22%
Avg. R&D intensity	5.47%	5.98%	6.09%	6.56%
Aggregate output	4.318	4.415	4.516	4.622
Avg. merger probability	1.56%	1.85%	1.63%	1.94%
Consumption	1.940	1.991	2.020	2.077
Avg. firm growth rate	8.23%	8.48%	8.57%	8.80%

This table reports model implications for the two subsamples partitioned based on information asymmetry. For each subsample estimation, we compare the baseline model and the counterfactual economy in which information frictions are eliminated by setting $\omega=0$. Columns 1 and 3 show the baseline results for the subsamples with low and high RMSEs of analysts' forecast errors, respectively. Columns 2 and 4 report the long-term effect of eliminating information frictions in which we allow firm distribution to evolve endogenously and let firms reoptimize their innovation policies. Subsamples are partitioned as described in Table 9, and Table 6 defines the variables.

Overall, the subsample estimation results confirm that our estimated parameter ω can indeed pick up the underlying information frictions: for firms with higher ex ante information asymmetry, the acquirer's due diligence reveals only one quarter of the target's private information.

6.2 Extended model with target-side lack of information

A key assumption in our baseline model is that target firms are able to observe the realization of their contemporaneous innovation outcome. One possible concern is that even target firms, sometimes, may not know the realization of their ongoing innovation exactly. To gauge how this assumption affects our results, we extend the baseline model to allow for the possibility that the target has no private information and we explore how it affects firms' M&A and innovation decisions.

In the baseline model, there were two scenarios to consider: the case with imperfect information, where the target knows its own contemporaneous productivity, but the acquirer does not have access to this information; and the case with perfect information, where both the target and the acquirer know the contemporaneous productivity of both firms. In the extended model, we add a third scenario: a case in which neither the acquirer nor the target knows the contemporaneous productivity of the target firm, and thus both firms have to use rational expectations to formulate their optimal strategies. Let $\Upsilon \in [0, 1]$ denote the probability of this new scenario. The model timeline remains similar to that in the baseline, and the Bellman equations are updated to incorporate three scenarios: with probability $(1-\omega)(1-\Upsilon)$, the target's contemporaneous productivity is revealed to the acquirer; with probability $\omega(1-\Upsilon)$, it remains

the private information of the target; and with probability γ , neither the acquirer nor the target knows the target's contemporaneous productivity. The Internet Appendix C presents the details of this model extension.

We perform a set of robustness checks with different values of γ . Specifically, we first use the micro-level data on patent renewals provided by the USPTO to potentially uncover the possible value of γ , and then we show that the results remain largely robust in a range of γ that is empirically relevant.

In the United States, inventing firms have to incur large costs to file their patents in the form of application and patent attorney fees on top of the research and development costs to come up with the invention in the first place. Therefore, they only apply for a patent if the expected value of patenting the innovation is above the application costs. Once a patent is granted, firms must also pay a patent maintenance fee in the 4th, 8th, and 12th years after the grant date to maintain the patent. Unlike the R&D and application costs, patent maintenance fees are quite trivial. However, we observe that some firms choose not to renew their patents despite the trivial maintenance fees. When such an event is observed, we can infer that the firm revised its own valuation of the patent and decided that its value is below the expectation when the application was filed. The administrative data on patent renewals are thus informative of the frequency of events where a patent turns out to be a "dud", that is, patents that were initially thought to be worthwhile but turned out to be of insignificant value *ex post*. The frequency of patents that are not renewed in their fourth year is 9.15% for the patents granted to the sample of firms we study. For robustness, we consider low, medium, high, and very high lack of information scenarios, which correspond to $\gamma = \frac{9.15\%}{2}$, $\gamma = 9.15\%$, $\gamma = 2 \times 9.15\%$, and $\gamma = 3 \times 9.15\%$, respectively.

Because of space constraints, we report the results predicted by the extended model with different values of γ in Tables E5 to E8 in the Internet Appendix. A higher value of γ reduces information asymmetry, and we observe that the quantitative importance of information frictions is gradually diminished. However, even if we assume that target firms do not have private information regarding the realization of their contemporaneous innovation in almost 30% of cases, information frictions still play a crucial role. Specifically, as we set γ to 27.44% (i.e., three times as large as its calibrated value), eliminating information asymmetry (i.e., set $\omega = 0$) would increase firms' capitalized expected gains from M&A by 36% ($36\% = \frac{1.017}{0.749} - 1$).

These results suggest that the effect of information asymmetry and the resultant adverse selection risk indeed diminish as target firms become less informed of their own innovation outcomes. The inefficiencies induced by information frictions, however, remain quite substantial in a range of γ that is empirically relevant. Admittedly, jointly estimating the parameters γ and ω is beyond the scope of this paper but doing so remains an important challenge to entertain in future research.

7. Conclusions

Interactions between innovation and the market of M&A have been well studied in the literature. An active M&A market incentivizes some firms to specialize in innovation with the anticipation of being acquired in the future. Acquiring innovation, however, can be challenging because information frictions make it difficult to assess the value and impact of innovation, especially breakthrough innovation. In this paper, we document a robust inverted-U relationship between firm innovativeness and takeover exposure, a positive association between equity usage and target innovativeness, as well as a negative association between deal completion rate and target innovativeness.

We develop and estimate a general equilibrium model of acquiring innovation with endogenous productivity growth to quantify the effect of information frictions. We find substantial information frictions between acquirers and targets, and eliminating information frictions can increase firms' expected gains from the M&A market by 59%. A more efficient M&A market stimulates more firm innovation, resulting in higher firm productivity growth and business dynamism, and improves aggregate output and social welfare.

These findings highlight the importance of the M&A market as a channel of optimal resource allocation and technological progress. Policies that can alleviate the friction in this market would not only increase the number of mergers but also boost innovation across firm boundaries as firms innovate more in anticipation of future merger opportunities. At the same time, larger firms, which are relatively inefficient in R&D, can reduce their in-house R&D expenditure, making each dollar spent on innovation more effective. Business dynamism also improves, thanks to the higher turnover of firms, and all firms, regardless of size and innovativeness, experience an increase in firm value. Given these significant positive effects, we believe future analyses targeted toward uncovering new ways to reduce information frictions to be socially worthwhile.

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